YEAR 12 MATHEMATICS EXTENSION 1

Year 12 Chapter 17 Binomial Distributions

Lesson and Homework Pack

Revision	The normal distribution	Year 12 Advanced · Ch 16
Lesson 17A	Binomial probability	Past papers: 21 marks · 30 min
Lesson 17B	Binomial distributions	Past papers: 21 marks · 30 min
Lesson 17C	Normal approximations to a binomial	Past papers: 18 marks · 25 min
Lesson 17D	Sample proportions	Past papers: 19 marks · 30 min
Review	Chapter 17 review exercise	Mixed practice

Each lesson runs in the same order: **concept** → **tutor demo** → **your attempt** → **worked example** to keep for homework. Then do the **textbook homework** and finish with the **past-paper set** under exam conditions. The standard normal table is on the last page of the Revision section. Answers and full past-paper solutions are in the separate Homework Answers booklet.

CHAPTER REVISION

The normal distribution

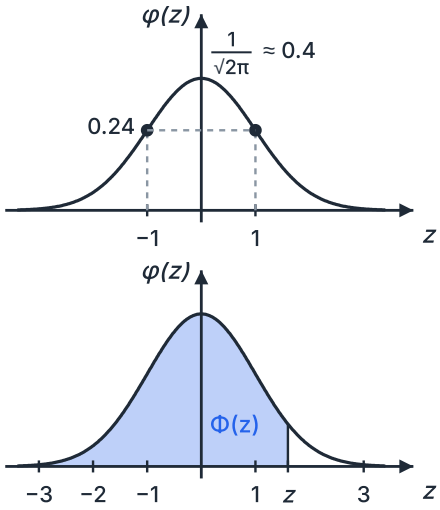
A short review of Year 12 Advanced Chapter 16D–16F: the standard normal distribution and its table, general normal distributions and z-scores, and applications. Lessons 17C and 17D approximate binomial distributions and sample proportions by normal distributions, so they use all of this. The standard normal table is on the last page of this section.

SUMMARY · keep these pages open while you work

THE STANDARD NORMAL DISTRIBUTION (16D)

Let Z be the **standard normal random variable**. Its probability density function is

$$\phi(z) = \frac{e^{-\frac{1}{2}z^2}}{\sqrt{2\pi}} \qquad \left(\frac{1}{\sqrt{2\pi}} \doteq 0.4\right)$$



- The graph of the PDF is a **bell-shaped curve**, with global maximum at $z = 0$ (the mode) and points of inflection at $z = 1$ and $z = -1$.
- The mean is $\mu = 0$ and the standard deviation is $\sigma = 1$. The points of inflection are each one standard deviation from the mean.
- The total area under the curve $y = \phi(z)$ is 1, and the curve is **even**: it has line symmetry in the y-axis.
- The **cumulative distribution function** is $\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(t)\,dt$, the area under the curve to the left of z. The primitive of $\phi(z)$ cannot be written with our usual functions, so we read $\Phi(z)$ from the **standard normal table** (last page of this section), or use a calculator.
- **Reading the table:** go to the row for the units and first decimal place of z, then to the column for the second decimal place. For example, $\Phi(1.63)$ is in row 1.6, column $+.03$, so $\Phi(1.63) \doteq 0.9484$. The table gives only $z \geq 0$, because $\phi(z)$ is even.

Rules for juggling areas (always sketch the curve and shade the area you want):

$$P(Z = a) = 0 \qquad \Phi(0) = 0.5 \qquad P(Z \geq z) = 1 - \Phi(z)$$
$$\Phi(-z) = P(Z \geq z) = 1 - \Phi(z) \qquad P(a \leq Z \leq b) = \Phi(b) - \Phi(a)$$

- Because $P(Z = a) = 0$, there is no need to distinguish between $\leq$ and $<$, or between $\geq$ and $>$.

WORKED EXAMPLE **Textbook Example 10 (16D)**

a Look up $P(Z \leq 2)$ and illustrate it as an area under $y = \phi(z)$.
b Illustrate each probability as an area under $y = \phi(z)$. Then calculate it using the value of $\Phi(2)$ found in part **a**. Keep looking back to the graph in part **a** while you juggle the intervals.

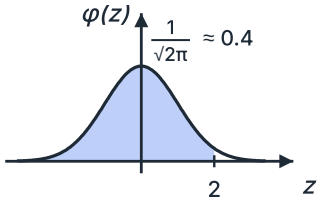
i $P(Z \geq 2)$ ii $P(Z \leq -2)$ iii $P(0 \leq Z \leq 2)$ iv $P(-2 \leq Z \leq 2)$

SOLUTION

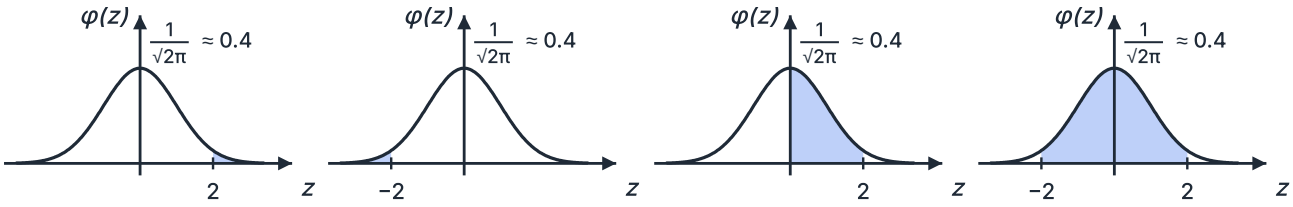
a From the table,

$$P(Z \leq 2) = \Phi(2) = 0.9772.$$

The area under $y = \phi(z)$ corresponding to $\Phi(2)$ is shaded in the diagram to the right.



b



i $P(Z \geq 2)$ ii $P(Z \leq -2)$ iii $P(0 \leq Z \leq 2)$ iv $P(-2 \leq Z \leq 2)$

i $P(Z \geq 2) = 1 - P(Z \leq 2)$, because the total area is 1, $\doteq 1 - 0.9772 \doteq 0.0228.$
The probability that $Z = 2$ exactly is zero, so there is no need to distinguish between $\leq$ and $<$ or between $\geq$ and $>$.

ii $P(Z \leq -2) = P(Z \geq 2)$, because $\phi(z)$ is even, $\doteq 0.0228$, from part i.

iii $P(0 \leq Z \leq 2) = \Phi(2) - \Phi(0)$, using subtraction of areas,
$\doteq 0.9772 - 0.5$, because exactly half the scores are below the mean,
$\doteq 0.4772.$

iv $P(-2 \leq Z \leq 2) = \Phi(2) - \Phi(-2) \doteq 0.9772 - 0.0228$, by part ii, $\doteq 0.9544$,
OR $P(-2 \leq Z \leq 2) = 2 \times P(0 \leq Z \leq 2)$, by symmetry, $\doteq 2 \times 0.4772$, by part iii, $\doteq 0.9544.$

WORKED EXAMPLE **More symmetry: intervals that are not centred on 0**

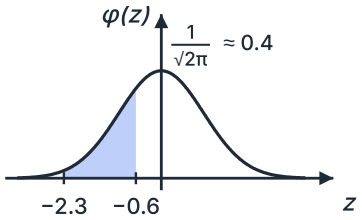
Use symmetry and the standard normal table to find:

a $P(-2.3 \leq Z \leq -0.6)$ b $P(-1.4 \leq Z \leq 0.9)$

SOLUTION

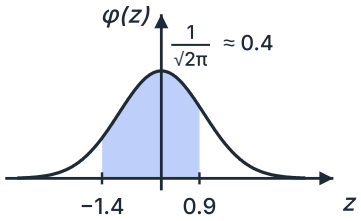
a Both ends are negative. Reflect in the y-axis, because $\phi(z)$ is even:

$$\begin{aligned} P(-2.3 \leq Z \leq -0.6) &= P(0.6 \leq Z \leq 2.3) \\ &= \Phi(2.3) - \Phi(0.6) \\ &\doteq 0.9893 - 0.7257 \\ &\doteq 0.2636. \end{aligned}$$



b The interval straddles 0, with unequal ends. Use $\Phi(-1.4) = 1 - \Phi(1.4)$:

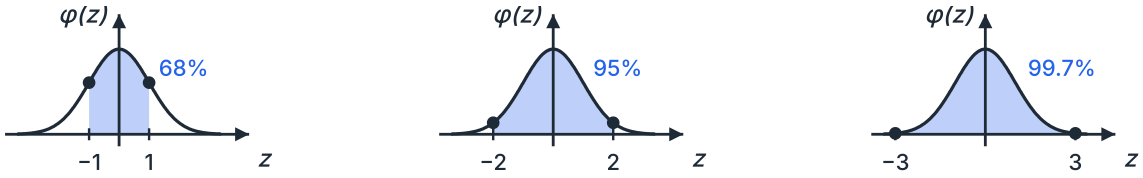
$$\begin{aligned} P(-1.4 \leq Z \leq 0.9) &= \Phi(0.9) - \Phi(-1.4) \\ &= \Phi(0.9) - \big(1 - \Phi(1.4)\big) \\ &= \Phi(0.9) + \Phi(1.4) - 1 \\ &\doteq 0.8159 + 0.9192 - 1 \\ &\doteq 0.7351. \end{aligned}$$



THE EMPIRICAL RULE, AND READING THE TABLE IN REVERSE (16D)

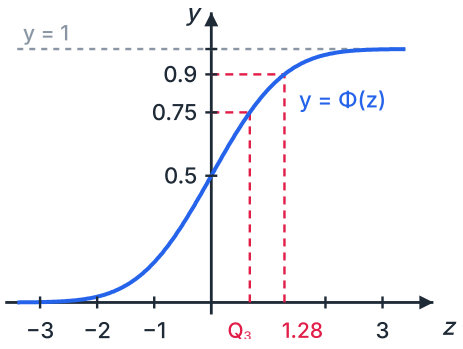
From the table, $P(-1 \leq Z \leq 1) \doteq 0.6827 \doteq 68\%$, $P(-2 \leq Z \leq 2) \doteq 0.9545 \doteq 95\%$ and $P(-3 \leq Z \leq 3) \doteq 0.9973 \doteq 99.7\%$. These are the **empirical rule**, or the **68–95–99.7 rule**. In a normal distribution, the proportion of scores lying:

- within 1 standard deviation of the mean is 68%,
- within 2 standard deviations of the mean is 95%,
- within 3 standard deviations of the mean is 99.7%.



Use the symmetry of the curve with these: for example, $P(Z > 1) \doteq \frac{1}{2}(100\% - 68\%) = 16\%$ and $P(0 \leq Z \leq 2) \doteq \frac{1}{2} \times 95\% = 47.5\%$.

Reading the table in reverse. To solve $\Phi(z) = p$, find p in the body of the table and read off z. Take the closest entry, or interpolate between two entries. The graph of $y = \Phi(z)$ on the right shows this as a horizontal line at height p. If $p < 0.5$, then z is negative: use $\Phi(-z) = 1 - p$.



It is standard practice to round percentiles and quartiles of the normal to two decimal places. The quartiles are $Q_1 \doteq -0.67$ and $Q_3 \doteq 0.67$, and $\text{IQR} \doteq 1.35$. By the IQR criterion, an outlier is a score more than $1.5 \times \text{IQR}$ beyond a quartile, that is, outside $-2.70 \leq z \leq 2.70$.

WORKED EXAMPLE **Textbook Example 13 (16D)**

a Find the 9th decile of the standard normal distribution.
b Find the third quartile Q_3 and the first quartile Q_1 of $\Phi(z)$.
c Using the IQR criterion, what proportion of the scores of a standard normal random variable would be expected to be outliers?

SOLUTION

a For the 9th decile, we need to solve $\Phi(z) \doteq 0.9$.
From the table, $\Phi(1.2) \doteq 0.8849$ and $\Phi(1.3) \doteq 0.9032$, with difference 0.0183 making the 9th decile about 1.28.
This agrees with the horizontal line with height 0.9 on the graph of $\Phi(z)$.

b For the upper quartile, we need to solve $\Phi(z) \doteq 0.75$.
From the table, $\Phi(0.6) \doteq 0.7257$ and $\Phi(0.7) \doteq 0.7580$, with difference 0.0323.
Thus $Q_3 \doteq 0.675$, and because $\phi(z)$ is even, $Q_1 \doteq -0.675$.
This agrees with the horizontal line with height 0.75 on the graph of $\Phi(z)$.

c From part **b**, the IQR is about 1.35, so $Q_3 + 1.5 \times \text{IQR} \doteq 2.70$.
The IQR criterion is that an outlier lies outside $-2.70 \leq z \leq 2.70$.

$$\begin{aligned} \textbf{Hence} \quad P(Z \text{ is an outlier}) &= P(Z > 2.70) + P(Z < -2.70) \\ &= 2 \times P(Z > 2.70) \\ &= 2\big(1 - \Phi(2.70)\big) \\ &\doteq 0.007, \end{aligned}$$

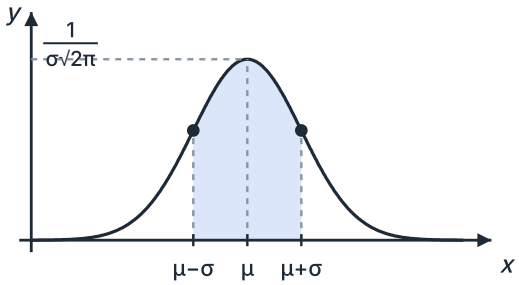
so roughly 7 in 1000 scores would be expected to be outliers.

Note: Correct to five significant figures, $Q_3 \doteq 0.67449$. It is standard practice to round percentiles and quartiles of the normal to two decimal places, giving $Q_1 \doteq -0.67$, $Q_3 \doteq 0.67$ and $\text{IQR} \doteq 1.35$. In a normal distribution, the IQR criterion makes just under 1% of scores outliers.

With the two-decimal table you can read these directly: $\Phi(1.28) \doteq 0.8997$ is the entry closest to 0.9, so the 9th decile is $z \doteq 1.28$, and $\Phi(0.67) \doteq 0.7486$ is the entry closest to 0.75, so $Q_3 \doteq 0.67$.

GENERAL NORMAL DISTRIBUTIONS AND z-SCORES (16E)

Let $f(x)$ be the PDF of a normal distribution with mean μ and standard deviation σ. We write $X \sim N(\mu, \sigma^2)$.



- $f(x)$ is obtained from the standard normal PDF by stretching horizontally with factor σ and vertically with factor $\dfrac{1}{\sigma}$, then shifting right by μ units:

$$f(x) = \frac{1}{\sigma}\,\phi\!\left(\frac{x-\mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}}\,e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- The points of inflection are each one standard deviation from the mean, at $x = \mu - \sigma$ and $x = \mu + \sigma$. The mean, the median and the mode coincide.
- The **z-score** of a score x is the number of standard deviations that x lies above the mean (negative if x is below the mean). The conversions are

$$z = \frac{x-\mu}{\sigma} \qquad \textbf{and} \qquad x = \mu + \sigma z.$$

- If X is normal, then $Z = \dfrac{X-\mu}{\sigma}$ is standard normal, so $P(X \le a) = P\!\left(Z \le \dfrac{a-\mu}{\sigma}\right) = \Phi\!\left(\dfrac{a-\mu}{\sigma}\right)$. This is called **standardising**.
- The empirical rule becomes: 68% of scores lie in $\mu - \sigma \le x \le \mu + \sigma$, 95% in $\mu - 2\sigma \le x \le \mu + 2\sigma$, and 99.7% in $\mu - 3\sigma \le x \le \mu + 3\sigma$.

WORKED EXAMPLE **Textbook Example 15 (16E)**

A normally distributed random variable X has mean 100 and standard deviation 20.
a Write down the two conversion formulae between z-scores and values of x.
b Find: i $P(X \leq 110)$ ii $P(X \geq 90)$
c Find (nearest whole number) the value of a such that $P(X \leq a) = 0.98$.

SOLUTION

a $z = \dfrac{x - 100}{20}$ and $x = 100 + 20z$.

b i $P(X \leq 110)$
$= P(Z \leq 0.5)$
$= 0.69$

ii $P(X \geq 90) = P(Z \geq -0.5)$
$= P(Z \leq 0.5)$ $(\phi(x)$ is even$)$
$= 0.69$

c From the table, $\Phi(2.0) = 0.9772$ and $\Phi(2.1) = 0.9821$,
so by interpolation, $\Phi(2.06) = 0.98$.
Converting back to x-values, $a = 100 + 20 \times 2.06 = 141$.

With the two-decimal table you can read z directly: $\Phi(2.05) \doteq 0.9798$ is the entry closest to 0.98, so $z \doteq 2.05$ and $a \doteq 100 + 20 \times 2.05 = 141$, the same answer.

APPLICATIONS OF THE NORMAL DISTRIBUTION (16F)

- Identify μ and σ, and convert each score to a z-score. Sketch the curve and shade the region you want.
- If the z-scores are whole numbers of standard deviations, the empirical rule is often enough. Otherwise use the table (or technology).
- A percentage of a population is a probability times 100%. The expected number of scores in a region is (number of scores) $\times$ (probability).
- To find a score from a probability, read the table in reverse to find z, then convert back with $x = \mu + \sigma z$.

WORKED EXAMPLE **Textbook Example 17 (16F)**

The Happytime Chocolate Company manufactures a 100 g chocolate–nougat bar. As with any manufacturing process, these chocolate bars do not all have precisely the same weight, and these bars are known to be normally distributed with standard deviation 2 g. To reduce the number of complaints, the company has adjusted its machinery so that the mean is 102 g (such an adjustment does not affect the standard deviation).

a Using the empirical rule where possible, and tables or technology otherwise, find the percentage of chocolate bars:

i of weight less than the stated weight of 100 g,

ii of weight greater than 105 g.

b What would the mean weight need to be for there to be less than 1 chocolate bar in 1000 under 100 g?

SOLUTION

a i A weight of 100 g is 1 standard deviation below the mean of 102 g.

By the empirical rule, $P(-1 \leq Z \leq 1) = 68\%$,
and using the complement, $P(Z < -1 \text{ or } Z > 1) = 32\%$,
so by the even symmetry, $P(Z < -1) = 32\% \div 2 = 16\%$.

ii A weight of 105 g is 1.5 standard deviations above the mean of 102 g.

From the table, $P(Z \leq 1.5) = 93\%$,
so using complements, $P(Z > 1.5) = 7\%$.

b Reading the table backwards, $P(Z \leq 3.1) = 0.999$,
so by symmetry and complements, $P(Z \leq -3.1) = 0.001$.
Hence we need to make the mean 3.1σ above 100 g,
meaning that we set the controls so that $\mu = 100 + 3.1 \times 2 = 106.2$ g.

WARM-UP · answers are in the answers booklet

Use the standard normal table at the end of this section. Each question is from the textbook exercise named, with some parts left out.

1 Exercise 16D Q9

Use the standard normal table to find:

a $P(Z \leq 1.3)$

b $P(Z = 2.4)$

c $P(Z > 0.4)$

d $P(Z \leq -0.8)$

e $P(1.1 \leq Z \leq 1.5)$

f $P(-1.3 \leq Z \leq 2.2)$

2 Exercise 16D Q12

Use the empirical rule (also called the 68–95–99.7 rule) to find:

a $P(Z \leq 1)$

b $P(Z < -1)$

c $P(0 \leq Z \leq 3)$

d $P(-3 < Z \leq -2)$

e $P(-2 \leq Z < 1)$

3 Exercise 16D Q14

Use the standard normal table in reverse to find the value of a, given that:

a $P(Z < a) = 0.7257$

b $P(Z < -a) = 0.1151$

c $P(Z < a) = 0.2119$

d $P(-a \leq Z < a) = 0.7286$

4 Exercise 16E Q6

Find each probability for a normally distributed random variable X with the given parameters. You will need to use the table of values for the standard normal distribution, or a statistics calculator, or other technology such as a spreadsheet, or online resources.

a $P(3 \leq X \leq 7)$, given mean $\mu = 5$ and standard deviation $\sigma = 0.8$.

b $P(X \leq 8)$, where $\mu = 12$ and $\sigma = 5$.

c $P(X \geq -39)$, where $\mu = 0$ and $\sigma = 30$.

d $P(3 < X \leq 5)$, where $\mu = 8$ and $\sigma = 2$.

5 Exercise 16F Q2 (use the empirical rule)

The results in an examination are approximately normally distributed with mean 68 and standard deviation 9. In a cohort of 2000, how many students will be expected to score:

a more than 95,

b less than 50,

c between 59 and 86?

6 Exercise 16F Q7

In Australian adult males, height is found to be normally distributed with mean 176 cm and standard deviation 7.5 cm. A doorway is designed so that 90% of this population can enter without ducking.

a Read the supplied standard normal distribution table backwards to find the z-score such that $P(Z < z) = 90\%$, correct to 2 decimal places.

b Hence find the minimum height of the doorway.

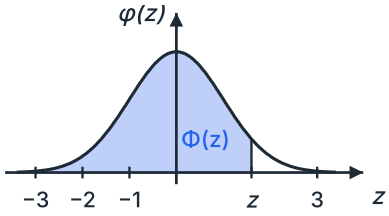
c In the Dinaric alps, the mean and standard deviation of the heights of adult males are respectively 185 and 7.5 centimetres. A customer orders a special design for the doorway so that 95% of adult males can enter without ducking.

i Explain why a reasonable estimate from the table such that $P(Z < z) = 0.95$ is 1.65.

ii Find the minimum design height of the door.

STANDARD NORMAL TABLE · values of $\Phi(z) = P(Z \leq z)$

The shaded area in the graph represents the value of the cumulative standard normal distribution function $P(Z \leq z) = \Phi(z) = \int_{-\infty}^{z} \phi(t)\,dt$. The table gives $\Phi(z)$ to four decimal places, allowing two decimal places for z. After that, use interpolation. For example,



$$P(Z \leq 1.627) = \Phi(1.627)$$
$$\doteq 0.9474 + 0.7\,(0.9484 - 0.9474) \doteq 0.9481.$$

For negative z, use $\Phi(-z) = 1 - \Phi(z)$. For $z \geq 4$, $\Phi(z) \doteq 1.0000$.

	second decimal place									
z	+.00	+.01	+.02	+.03	+.04	+.05	+.06	+.07	+.08	+.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

LESSON 17A

Binomial probability

This lesson is about individual binomial probabilities, in preparation for the later sections on binomial distributions. The experiments have a special form: the stages are independent, each stage has just two outcomes, 'success' and 'failure', with the same probabilities at each stage, and we record only the number of successes, not their order. The probability of x successes in n stages is ${}^nC_x\, p^x\, q^{n-x}$, a term of the binomial expansion of $(p+q)^n$. Complementary events and cases handle 'at least' and 'at most'.

CONCEPT 1 · Bernoulli trials and binomial experiments

BERNOULLI TRIALS

It is convenient to have a name for the independent stages and to describe them carefully. A *Bernoulli trial* or *Bernoulli experiment* is a single-stage random experiment with just two outcomes, conventionally called 'success' and 'failure', to which we usually assign the probabilities p and $q = 1 - p$.

- The classic example of a Bernoulli trial is tossing a coin, where 'success' is heads and 'failure' is tails. The probability of success is then $p = \frac{1}{2}$, and the probability of failure is $q = 1 - p = \frac{1}{2}$.
- The other classic example is throwing a die, provided that we define 'success'. If we define 'success' to be 'throwing a six', and record only that, then we have a Bernoulli trial with $p = \frac{1}{6}$.
- A Bernoulli trial is completely determined by just one parameter, the probability p of 'success'.

BINOMIAL EXPERIMENTS

A *binomial experiment* is an n-stage experiment in which:

- each stage is a Bernoulli trial with the same probability p of success,
- the stages are independent: no stage affects any other stage,
- the random variable X is the number of successes: order is irrelevant.

Think now about tossing a coin 12 times and counting the number of heads. Or think about throwing four dice and counting the number of sixes.

A binomial experiment is thus completely determined by just two parameters, the number n of trials, and the probability p of success at each stage.

Each stage of a binomial experiment is trivially a binomial experiment with just one stage. Thus a Bernoulli trial is a special case of a binomial experiment.

TUTOR DEMO **Bernoulli trial or not? Binomial or not?**

a Is each experiment a Bernoulli trial? If it is, say what 'success' is and give p.

(i) A die is thrown, and it is recorded whether it shows an even number.

(ii) A die is thrown, and the number showing is recorded.

(iii) A card is drawn from a pack of 52 cards, and it is recorded whether it is a heart.

b Is X the random variable of a binomial experiment? If it is, give n and p.

(i) A coin is tossed 10 times, and X is the number of heads.

(ii) A die is thrown until it shows a six, and X is the number of throws.

(iii) Three cards are drawn from a pack of 52 cards without replacement, and X is the number of hearts.

YOUR TURN **Same skill**

a Is each experiment a Bernoulli trial? If it is, say what 'success' is and give p.

(i) A spinner with five equal sectors numbered 1 to 5 is spun, and it is recorded whether it lands on 5.

(ii) A student is chosen at random, and his or her height is recorded.

(iii) A die is thrown, and it is recorded whether it shows a number greater than 4.

b Is X the random variable of a binomial experiment? If it is, give n and p.

(i) A die is thrown 8 times, and X is the number of threes.

(ii) Two balls are drawn without replacement from a bag with 3 red and 2 blue balls, and X is the number of red balls.

(iii) A die is thrown 6 times, and X is the total of the numbers shown.

WORKED EXAMPLE 1 **Keep for homework**

Identify some further examples of Bernoulli trials.

SOLUTION

- Choose an adult Australian. Did they vote in the last election?
- Choose a shopper in the street. Have they visited Wooldi today?
- Ask a whale-spotting boat returning to port: did you spot a whale?
- Visit your letter-box. Is there mail in it or not?
- Is there intelligent life elsewhere in the universe? Yes or no?
- Choose a person at random. Were they offered the last job they applied for?

CONCEPT 2 · repeatedly attempting to throw a six

INDIVIDUAL PROBABILITIES IN A BINOMIAL EXPERIMENT

Here is a classic example. A die is thrown four times. Find the probabilities of getting 0, 1, 2, 3 or 4 sixes. Let S (success) be 'throwing a six' and F (failure) be 'not throwing a six', so that $p = \frac{1}{6}$ and $q = 1 - p = \frac{5}{6}$.

- A probability tree diagram shows the sixteen possible outcomes taking account of order, and their probabilities (see the worked example below).
- When we ignore order, these 16 outcomes collapse to five, and the resulting **binomial random variable** X has five possible values, 0, 1, 2, 3 and 4.
- The outcome 'two sixes', for example, can be obtained in ${}^4C_2 = 6$ different ways, because there are $\frac{4!}{2! \times 2!} = {}^4C_2$ ways of arranging two Ss and two Fs. Each of these six outcomes has the same probability, so the probabilities add.
- The five probabilities are the successive terms in the binomial expansion of $\left(\frac{1}{6} + \frac{5}{6}\right)^4 = 1^4 = 1$, so they add to 1.

TUTOR DEMO **Three spins of a spinner**

A spinner lands on red with probability $\frac{1}{3}$. It is spun three times, and X is the number of times it lands on red.

a Draw a probability tree diagram showing the eight possible outcomes, taking account of order, and their probabilities.

b Write $P(X=0)$, $P(X=1)$, $P(X=2)$ and $P(X=3)$ using ${}^3C_x$, and find their values.

c Show that the four probabilities are the terms in the binomial expansion of $\left(\frac{1}{3}+\frac{2}{3}\right)^3$.

YOUR TURN **Same method**

A card is drawn at random from a pack of 52 playing cards, and it is recorded whether it is a heart. The card is replaced and the pack is shuffled. This is done three times, and X is the number of hearts.

a Draw a probability tree diagram showing the eight possible outcomes, taking account of order, and their probabilities.

b Write $P(X=0)$, $P(X=1)$, $P(X=2)$ and $P(X=3)$ using ${}^3C_x$, and find their values.

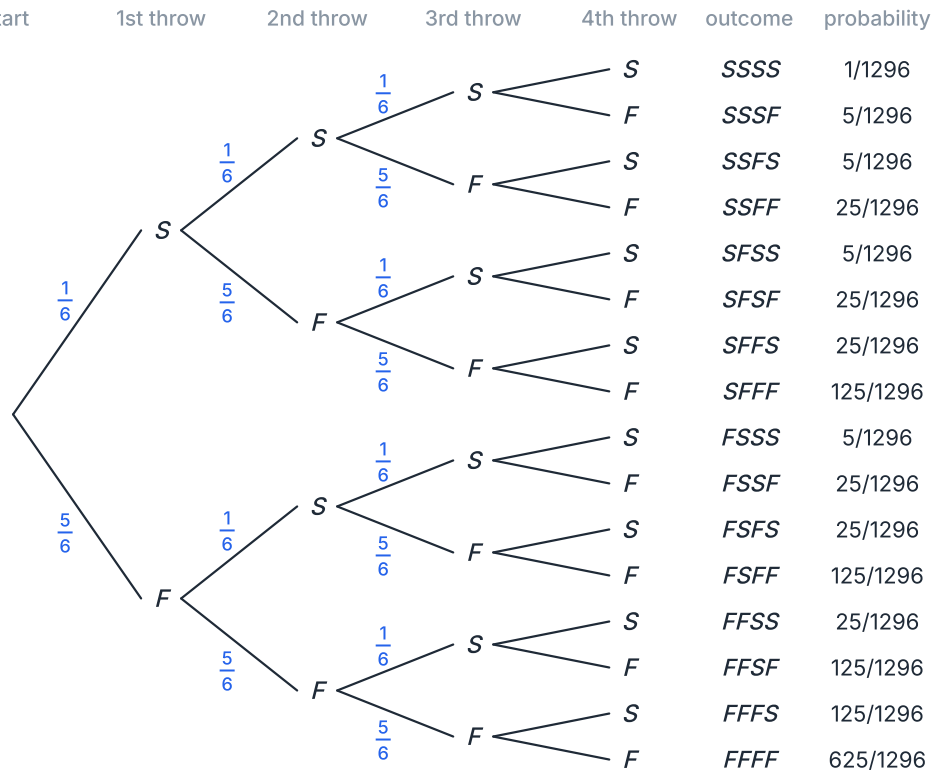
c Show that the four probabilities are the terms in the binomial expansion of $\left(\frac{1}{4}+\frac{3}{4}\right)^3$.

WORKED EXAMPLE **From the text: keep for homework**

A die is thrown four times. Find the probabilities of getting 0, 1, 2, 3 or 4 sixes.

SOLUTION

Let S (success) be 'throwing a six' and F (failure) be 'not throwing a six', so that $p = \frac{1}{6}$ and $q = 1 - p = \frac{5}{6}$. Here is the probability tree diagram showing the sixteen possible outcomes taking account of order, and their respective probabilities. Each branch to S has probability $\frac{1}{6}$ and each branch to F has probability $\frac{5}{6}$, so for example $P(\text{SSFF}) = \frac{1}{6} \times \frac{1}{6} \times \frac{5}{6} \times \frac{5}{6} = \frac{25}{1296}$.



When we ignore order, these 16 outcomes collapse to five, and the resulting binomial random variable X has five possible values, 0, 1, 2, 3 and 4.

The outcome 'two sixes', for example, can be obtained in ${}^4C_2 = 6$ different ways: SSFF, SFSF, SFFS, FSSF, FSFS, FFSS, because there are $\frac{4!}{2! \times 2!} = {}^4C_2$ ways of arranging two Ss and two Fs. Each of these six outcomes has the same probability $\left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^2$, so $P(X = 2) = {}^4C_2 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^2$.

Similar arguments apply to the probabilities of getting 0, 1, 3 and 4 sixes:

Result	Probability	Approximation
0 sixes	$^4C_0 \times \left(\frac{1}{6}\right)^0 \times \left(\frac{5}{6}\right)^4$	0.482 25
1 six	$^4C_1 \times \left(\frac{1}{6}\right)^1 \times \left(\frac{5}{6}\right)^3$	0.385 80
2 sixes	$^4C_2 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^2$	0.115 74
3 sixes	$^4C_3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^1$	0.015 43
4 sixes	$^4C_4 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)^0$	0.000 77
		Sum $= 1$

The five probabilities of course add up to 1, because no other outcomes are possible. This is also clear because the five probabilities are the successive terms in the binomial expansion of $\left(\frac{1}{6} + \frac{5}{6}\right)^4 = 1^4 = 1$:

$$\left(\frac{1}{6} + \frac{5}{6}\right)^4 = {}^4C_0 \times \left(\frac{1}{6}\right)^0 \times \left(\frac{5}{6}\right)^4 + {}^4C_1 \times \left(\frac{1}{6}\right)^1 \times \left(\frac{5}{6}\right)^3 + {}^4C_2 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^2 + {}^4C_3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{5}{6}\right)^1 + {}^4C_4 \times \left(\frac{1}{6}\right)^4 \times \left(\frac{5}{6}\right)^0.$$

Remember that

$$(x+y)^4 = {}^4C_0\, x^0 y^4 + {}^4C_1\, x^1 y^3 + {}^4C_2\, x^2 y^2 + {}^4C_3\, x^3 y^1 + {}^4C_4\, x^4 y^0.$$

CONCEPT 3 · binomial probability, the general case

BINOMIAL PROBABILITY: THE GENERAL CASE

Suppose that a multi-stage experiment consists of n identical stages, and at each stage the probability of 'success' is p and of 'failure' is q, where $p + q = 1$. Then

$$P(x \text{ successes and } n - x \text{ failures in that order}) = p^x\, q^{n-x}.$$

But there are ${}^nC_x$ ways of ordering x successes and $(n - x)$ failures, so

$$P(x \text{ successes and } n - x \text{ failures in any order}) = {}^nC_x\, p^x\, q^{n-x}.$$

This is the term in $p^x q^{n-x}$ in the expansion of the binomial $(p + q)^n$.

1 BERNOULLI TRIALS AND BINOMIAL PROBABILITY

- A *Bernoulli trial* or *Bernoulli experiment* is an experiment with two outcomes, 'success' and 'failure', usually assigned the probabilities p and $q = 1 - p$.
- A *binomial random variable* is the outcome of an n-stage experiment in which:
  each stage is a Bernoulli trial with the same probability p of success,
  the stages are independent,
  the random variable X is the number of successes, not their order.
- Suppose that the probabilities of 'success' and 'failure' in any stage of an n-stage binomial experiment are p and $q = 1 - p$ respectively. Then
  $P(x \text{ successes}) = {}^nC_x\, p^x\, q^{n-x}$.
- This probability is the term in $p^x q^{n-x}$ in the expansion of $(p+q)^n = 1^n = 1$, and the sum of the $n+1$ binomial probabilities is 1.
- In particular, if $p = q = \frac{1}{2}$, then the formula simplifies to
  $P(x \text{ successes}) = {}^nC_x \left(\frac{1}{2}\right)^n$.

NOTATION, AND 'AT LEAST' QUESTIONS

- The pronumeral x is usually used for the values of a discrete random variable. In this chapter we will therefore mostly use x in place of the r that was used in Chapters 14–15 of the Year 11 book.
- Complementary events and cases answer 'at least' and 'at most' questions. For example, $P(\text{at least one}) = 1 - P(\text{none})$, and
  $P(\text{at least four of six}) = P(4) + P(5) + P(6)$, adding the cases because they are mutually exclusive (see Worked example 2).

TUTOR DEMO **Seven throws of a die**

A die is thrown seven times. A throw is a ‘success’ if it shows a number less than 3. Find, as fractions with denominator 3^7, the probability that:

a exactly two throws are successes,

b exactly one throw is a success,

c at least one throw is a success,

d at least six throws are successes.

YOUR TURN **Same method**

Five cards are drawn at random from a pack of 52 playing cards. Each card is replaced and the pack is shuffled before the next card is drawn. Find, as fractions with denominator 4^5, the probability that:

a two are hearts,

b one is a heart,

c at least one is a heart,

d at least four are hearts.

WORKED EXAMPLE 2 **Keep for homework**

Six cards are drawn at random from a pack of 52 playing cards. Each card is replaced and the pack is shuffled before the next card is drawn. Find, as fractions with denominator 4^6, the probability that:

a two are clubs,

b one is a club,

c at least one is a club,

d at least four are clubs.

SOLUTION

There are 13 clubs in the pack, so at each stage the probability of drawing a club is $\frac{1}{4}$. Applying the formula with $p = \frac{1}{4}$ and $q = \frac{3}{4}$:

a

$$\begin{aligned} P(\text{two are clubs}) &= {}^6C_2 \times \left(\tfrac{1}{4}\right)^2 \times \left(\tfrac{3}{4}\right)^4 \\ &= \frac{15 \times 3^4}{4^6} \\ &= \frac{1215}{4^6} \end{aligned}$$

b

$$\begin{aligned} P(\text{one is a club}) &= {}^6C_1 \times \left(\tfrac{1}{4}\right)^1 \times \left(\tfrac{3}{4}\right)^5 \\ &= \frac{6 \times 3^5}{4^6} \\ &= \frac{1458}{4^6} \end{aligned}$$

c

$$\begin{aligned} P(\text{at least one is a club}) &= 1 - P(\text{all are non-clubs}) \\ &= 1 - \left(\tfrac{3}{4}\right)^6 \qquad \left(\text{or } 1 - {}^6C_0 \times \left(\tfrac{1}{4}\right)^0 \times \left(\tfrac{3}{4}\right)^6\right) \\ &= \frac{3367}{4^6} \end{aligned}$$

d

$$\begin{aligned} P(\text{at least four are clubs}) &= P(\text{four are clubs}) + P(\text{five are clubs}) \\ &\quad + P(\text{six are clubs}) \\ &= {}^6C_4 \times \left(\tfrac{1}{4}\right)^4 \times \left(\tfrac{3}{4}\right)^2 + {}^6C_5 \times \left(\tfrac{1}{4}\right)^5 \times \left(\tfrac{3}{4}\right)^1 \\ &\quad + {}^6C_6 \times \left(\tfrac{1}{4}\right)^6 \times \left(\tfrac{3}{4}\right)^0 \\ &= \frac{15 \times 3^2 + 6 \times 3 + 1}{4^6} \\ &= \frac{154}{4^6} \end{aligned}$$

CONCEPT 4 · an example where $p = q = \frac{1}{2}$

WHEN $p = q = \frac{1}{2}$

A particular case of binomial probability is when the probabilities p and q of 'success' and 'failure' are both $\frac{1}{2}$. Then $p^x q^{n-x} = \left(\frac{1}{2}\right)^n$ for every x, and as stated in Box 1,

$$P(x \text{ heads in } n \text{ tosses of a coin}) = {}^nC_x \times \left(\frac{1}{2}\right)^n.$$

TUTOR DEMO Twenty tosses

A coin is tossed 20 times. Find, correct to four significant figures, the probability that it comes up heads:
a exactly 10 times,
b exactly 12 times.

YOUR TURN Same method

A coin is tossed 30 times. Find, correct to four significant figures, the probability that it comes up heads:
a exactly 15 times,
b exactly 18 times.

WORKED EXAMPLE 3 **Keep for homework**

If a coin is tossed 100 times, what is the probability that it comes up heads exactly 50 times (correct to four significant figures)?

SOLUTION

Taking $p = q = \frac{1}{2}$,

$$\begin{aligned} P(\text{50 heads}) &= {}^{100}C_{50} \times \left(\tfrac{1}{2}\right)^{50} \times \left(\tfrac{1}{2}\right)^{50} \\ &= {}^{100}C_{50} \times \left(\tfrac{1}{2}\right)^{100} \\ &\doteq 0.07959. \end{aligned}$$

Note: This is a fairly low probability. Should we have expected a higher probability than this? Hardly, because any result from about 45 to 55 heads would be unlikely to surprise us. In general, as already stated in Box 1 above,

$$P(x \text{ heads in } n \text{ tosses of a coin}) = {}^{n}C_{x} \times \left(\tfrac{1}{2}\right)^{n}.$$

CONCEPT 5 · experimental probabilities and binomial probability

EXPERIMENTAL PROBABILITIES

Some of the most straightforward and important applications of binomial theory arise in situations where the probabilities of 'success' and 'failure' are determined experimentally. For example, a manufacturer may find by careful testing that 1% of its products are defective, and then use $p = 0.01$ for each item in a box (see Worked example 4).

HOW MANY TRIALS ARE NEEDED?

Some questions (for example Exercise 17A Q12) ask how many trials n are needed to make $P(\text{at least one success})$ larger than some number c. Use the complement: $P(\text{at least one success}) = 1 - q^n$. Then solve $1 - q^n > c$, that is, $q^n < 1 - c$, by taking logarithms. Since $0 < q < 1$, $\ln q$ is negative, so the inequality sign reverses when we divide by $\ln q$:

$$n \ln q < \ln(1 - c), \qquad n > \frac{\ln(1 - c)}{\ln q}.$$

The answer is the smallest whole number n that works. Check it, and the number before it.

TUTOR DEMO **Cracked eggs**

An egg farm finds, after careful testing, that 2% of its eggs are cracked. The eggs are packed in trays of 30.

a Find, correct to three significant figures, the probability that a tray contains no cracked eggs.

b Find, correct to three significant figures, the probability that at least two eggs in a tray are cracked.

c How many eggs must be checked so that the probability of finding at least one cracked egg is more than 0.9?

YOUR TURN **Same method**

A seed company finds, after careful testing, that 5% of its seeds fail to germinate. The seeds are sold in packets of 40.

a Find, correct to three significant figures, the probability that no seeds in a packet fail.

b Find, correct to three significant figures, the probability that at least two seeds in a packet fail.

c How many seeds must be planted so that the probability that at least one fails is more than 0.95?

WORKED EXAMPLE 4 **Keep for homework**

A light bulb is classed as 'defective' if it burns out in under 1000 hours. A company making light bulbs finds, after careful testing, that 1% of its bulbs are defective. If it packs its bulbs in boxes of 50, find, correct to three significant figures:
a the probability that a box contains no defective bulbs,
b the probability that at least two bulbs in a box are defective.

SOLUTION

In this case, $p = 0.01$ and $q = 0.99$. Let X be the number of defective bulbs in the box. Then:

a $P(X = 0) = 0.99^{50} \doteq 0.605$

b

$$\begin{aligned} P(X \geq 2) &= 1 - \big(P(X = 0) + P(X = 1)\big) \\ &= 1 - \big(0.99^{50} + {}^{50}C_1 \times 0.01^1 \times 0.99^{49}\big) \\ &\doteq 0.0894 \end{aligned}$$

CONCEPT 6 · each stage is a compound event

WHEN EACH STAGE IS A COMPOUND EVENT

Sometimes, when each stage of the experiment is itself a compound event, it may take some work to find the probability of success at each stage. Find p first, using the product rule and the addition rule (a tree diagram may help). Then apply the binomial formula with that p.

TUTOR DEMO **Two machines**

A factory has two machines. Machine A makes 60 bolts an hour, of which 3% are faulty. Machine B makes 40 bolts an hour, of which 5% are faulty. The bolts are mixed at random and packed in boxes of 20. Find, correct to three significant figures, the probability that no more than one bolt in a box is faulty.

YOUR TURN **Same method**

A bakery has two ovens. Oven 1 bakes 40 loaves an hour, of which 5% are burnt. Oven 2 bakes 60 loaves an hour, of which 10% are burnt. The loaves are mixed at random and packed in crates of 12. Find, correct to three significant figures, the probability that no more than one loaf in a crate is burnt.

WORKED EXAMPLE 5 **Keep for homework**

Joe King and his sister Fay make shirts for a living. Joe works more slowly, but more accurately, making 20 shirts a day, of which 2% are defective. Fay works faster, making 30 shirts a day, of which 4% are defective. If they send out their shirts in randomly mixed parcels of 30 shirts, what is the probability (correct to three significant figures) that no more than two shirts in a box are defective?

SOLUTION

If a shirt is chosen at random from one parcel, then using the product rule and the addition rule, the probability p that the shirt is defective is

$$\begin{aligned} p &= P(\text{Joe made it, and it is defective}) + P(\text{Fay made it, and it is defective}) \\ &= \frac{20}{50} \times \frac{2}{100} + \frac{30}{50} \times \frac{4}{100} \\ &= \frac{4}{125}, \end{aligned}$$

so $p = \frac{4}{125}$ and $q = \frac{121}{125}$. Let X be the number of defective shirts. Then

$$\begin{aligned} P(X \leq 2) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= \left(\tfrac{121}{125}\right)^{30} + {}^{30}C_1 \times \left(\tfrac{121}{125}\right)^{29} \times \tfrac{4}{125} + {}^{30}C_2 \times \left(\tfrac{121}{125}\right)^{28} \times \left(\tfrac{4}{125}\right)^2 \\ &\doteq 0.930. \end{aligned}$$

TEXTBOOK HOMEWORK · Exercise 17A

Unless otherwise specified, leave your answers in unsimplified form.

FOUNDATION

1 Assume that the probability that a child is female is $\frac{1}{2}$, that sex is independent from child to child, and that there are only two sexes. Giving your answers as fractions in simplest form, find the probability that in a family of five children:

a all are boys,

b there are two girls and three boys,

c there are four boys and one girl,

d at least one will be a girl.

2 In a one-day cricket game, a batsman has a chance of $\frac{1}{5}$ of hitting a boundary every time he faces a ball. If he faces all six balls in an over, what is the probability that he will hit exactly two boundaries, assuming that successive strikes are independent?

3 A jury roll contains 2000 names, 700 of females and 1300 of males. Twelve jurors are randomly selected.

a Explain why it is reasonable to make the approximation that the probability of selecting a male does not change with each selection.

b What is the probability of ending up with an all-male jury?

4 A die is rolled twelve times. Find the probability that 5 appears on the uppermost face:

a exactly three times,

b exactly eight times,

c ten or more times (that is ten, eleven or twelve times).

5 A die is rolled six times. Let N denote the number of times that the number 3 is shown on the uppermost face. Find, correct to four decimal places:

a $P(N = 2)$

b $P(N < 2)$

c $P(N \geq 2)$

6 An archer finds that on average he hits the bulls-eye nine times out of ten. Assuming that successive attempts are independent, find the probability that in twenty attempts:

a he scores at least eighteen hits,

b he misses at least once.

7 A torch manufacturer finds that on average 9% of the bulbs are defective. What is the probability that in a randomly selected batch of ten one-bulb torches:

a there will be no more than two with defective bulbs,

b there will be at least two with defective bulbs.

DEVELOPMENT

8 A coin is tossed four times and the result is recorded. Janice wins if there are exactly two heads.

a List all the ways that this could occur, that is, in any order.

b By counting, verify that there are exactly 6 cases where Janice wins and find the probability of this outcome.

c Explain why this is the same as the number of ways of ordering the word HHTT, and use combinatorics to count how many such arrangements exist. Does your answer agree with part **b**?

d Show that this is equivalent to choosing two of the coins and placing them heads up, while placing the other two coins tails up.

9 A poll indicates that 55% of people support the policies of the Working Together Party. If five people are selected at random, what is the probability that a majority of them will support WTP party policies? Give your answer correct to three decimal places.

10 The probability that a small earthquake occurs somewhere in the world on any one day is 0.95. Assuming that earthquake frequencies on successive days are independent (this assumption is probably false), what is the probability that a small earthquake occurs somewhere in the world on exactly 28 of January's 31 days? Leave your answer in index form.

11 The probability that a jackpot prize will be won in a given lottery is 0.012.

a Find, correct to five decimal places, the probability that the jackpot prize will be won:
(i) exactly once in ten independent lottery draws,
(ii) at least once in ten independent lottery draws.

b The jackpot prize is initially $\$10\,000$ and increases by $\$10\,000$ each time the prize is not won. Find, correct to five decimal places, the probability that the jackpot prize will exceed $\$200\,000$ when it is finally won.

12 **a** How many times must a die be rolled so that the probability of rolling at least one six is greater than 95%?

b How many times must a coin be tossed so that the probability of tossing at least one tail is greater than 99%?

13 Five families have three children each.

a Find, correct to three decimal places, the probability that:
(i) at least one of these families has three boys,
(ii) each family has more boys than girls.

b What assumptions have been made in arriving at your answer?

14 Comment on the validity of the following arguments:

a 'In the McLaughlin Library, 10% of the books are mathematics books. Hence if I go to a shelf and choose five books from that shelf, then the probability that all five books are mathematics books is 10^{-5}.'

b 'During an election, 45% of voters voted for party A. Hence if I select a street at random, and then select a voter from each of four houses in the street, the probability that exactly two of those voters voted for party A is ${}^4C_2 \times (0.45)^2 \times (0.55)^2$.'

15 During winter it rains on average 18 out of 30 days. Five winter days are selected at random. Find, correct to four decimal places, the probability that:

a the first two days chosen will be fine and the remainder wet,

b more rainy days than fine days have been chosen.

16 A tennis player finds that, on average, he gets his serve in in eight out of every ten attempts, and that he serves an ace (a serve which is in the boundaries and not touched by his opponent) once every fifteen serves. He serves four times. Assuming that successive serves are independent events, find, correct to six decimal places, the probability that:

a all four serves are in,

b he hits at least three aces,

c he hits exactly three aces and the other serve lands in.

17 A man is restoring ten old cars, six of them manufactured in 1955 and four of them manufactured in 1962. When he tries to start them, on average the 1955 models will start 65% of the time and the 1962 models will start 80% of the time. Find, correct to four decimal places, the probability that at any time:

a exactly three of the 1955 models and one of the 1962 models will start,

b exactly four of the cars will start. (Hint: You will need to consider five cases.)

18 An apple exporter deals in two types of apples, Red Delicious and Golden Delicious. The ratio of Red Delicious to Golden Delicious is $4:1$. The apples are randomly mixed together before they are boxed. One in every fifty Golden Delicious and one in every one hundred Red Delicious apples will need to be discarded because they are undersized.

a What is the probability that an apple selected from a box will need to be discarded?

b If ten apples are randomly selected from a box, find the probability that:
(i) all the apples will have to be discarded,
(ii) half of the apples will have to be discarded,
(iii) less than two apples will be discarded.

19 One bag contains three red and five white balls, and another bag contains four red and four white balls.

a One bag is chosen at random, a ball is selected from that bag, its colour is noted, and then it is replaced. Find the probability that the ball chosen is red.

b If the operation in part **a** is carried out eight times, find the probability that:
(i) exactly three red balls are drawn,
(ii) at least three red balls are drawn.

20 **a** If six dice are rolled one hundred times, how many times would you expect the number of even numbers showing to exceed the number of odd numbers showing?

b If eight coins are tossed sixty times, how many times would you expect the number of heads to exceed the number of tails?

ENRICHMENT

21 A game is played using a barrel containing twenty similar balls numbered 1 to 20. The game consists of drawing four balls, without replacement, from the twenty balls in the barrel. Thus the probability that any particular number is drawn in any game is 0.2.

a Find as a decimal the probability that the number 19 is drawn in exactly two of the next five games played.

b Find as a decimal the probability that the number 19 is drawn in at least two of the next five games played.

c Let n be an integer, where $4 \leq n \leq 20$.
(i) What is the probability that, in any one game, all four selected numbers are less than or equal to n?
(ii) Show that the probability that, in any one game, n is the largest of the four numbers drawn is $\frac{{}^{n-1}C_3}{{}^{20}C_4}$.

22 **a** Expand $(a+b+c)^3$.

b In a survey of football supporters, 65% supported Hawthorn, 24% followed Collingwood and 11% followed Sydney. Use the expansion in part **a** to find, correct to five decimal places, the probability that if three people are randomly selected:
(i) one supports Hawthorn, one supports Collingwood and one supports Sydney,
(ii) exactly two of them support Collingwood,
(iii) at least two of them support the same team.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Northern Beaches Secondary College Manly Selective Campus, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q2** 1 mark

Which one of the following describes a Bernoulli random variable?

A. The times achieved by athletes in a 100-metre race

B. The value of a card chosen from a pack of 52 playing cards

C. Choosing a red pen from a container that contains red, black and blue pens

D. The number of emails a doctor receives in a week

Hint: how many outcomes does a Bernoulli trial have?

PP 2 **Blacktown Boys' High School, 2024, HSC Trial Examination, Mathematics Extension 1, Q6** 1 mark

Mohammad hits the target on average 2 out of every 3 shots in archery competitions. During a competition he has 10 shots at the target. What is the probability that Mohammad hits the target exactly 9 times?

A. $10\left(\frac{2}{3}\right)\left(\frac{1}{3}\right)^9$

B. $\left(\frac{2}{3}\right)^9\left(\frac{1}{3}\right)$

C. $10\left(\frac{2}{3}\right)^9$

D. $5\left(\frac{2}{3}\right)^{10}$

Hint: what are n and p here?

PP 3 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(c)** 2 marks

Five regular dice are rolled. Calculate the probability that exactly two of the dice show a 6 on the uppermost face.

Hint: what is a 'success' here, and what is its probability?

PP 4 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)** **2 marks**

A test consists of five multiple choice questions. Each question has four alternative answers. For each question only one of the alternative answers is correct. Jerry randomly selects an answer to each of the five questions.

(i) What is the probability that Jerry selects three correct and two incorrect answers? 2

Hint: which binomial distribution is this?

PP 5 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q13(b)(i)(ii)** **2 marks**

A toy car factory knows that 3% of the toy cars it produces are faulty. Toy cars are supplied in boxes of 50 cars, and boxes are supplied in pallets of 80 boxes.
Find the probability, to 4 decimal places, that:

(i) A box of toy cars contains exactly 5 faulty cars. 1

(ii) A box of toy cars contains at least 1 faulty car. 1

Hint: what are n and p here?

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q11(d)** 3 marks

In a particular racing car, the probability that the brake pads will need replacing during a race is 0.2. The car competes in a 12 race series.

(i) What is the probability that the pads will need to be replaced on exactly 3 occasions? Give your answer correct to 2 decimal places. 1

(ii) What is the probability that the pads will need to be replaced on 3 occasions at most? Give your answer correct to 2 decimal places. 2

Hint: which binomial distribution is this?

PP 7 **Blacktown Boys' High School, 2020, HSC Trial Examination, Mathematics Extension 1, Q12(a)** 3 marks

Bags of lollipops are supposed to contain 50 lollipops. Production records indicate that 92% of bags contain 50 lollipops. A batch of 15 bags is sampled. If more than 2 bags do not contain exactly 50 lollipops, production is stopped.

(i) Find the probability, correct to four decimal places, that exactly 2 of the 15 bags selected do not contain 50 lollipops. 1

(ii) Find the probability, correct to four decimal places, that production is stopped. 2

Hint: what counts as a 'success' here?

PP 8 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q11(b)** 3 marks

In a dice game, success is defined as obtaining a total of 9 when throwing two dice. Six rounds of the game are played. Find the probability of at least five successes. Give your answer to 3 significant figures.

Hint: what is p for one round?

PP 9 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)(ii)** 3 marks

Tomatoes are considered to be either determinate or indeterminate.
Mario buys tomatoes from his local store where the tomatoes are twice as likely to be sourced from Farm A than Farm B.
It is known that 60% of Farm A's tomatoes are determinate while 70% of Farm B's tomatoes are determinate. Mario cannot tell the difference between these tomatoes from their appearance.

(i) Show that the probability that a randomly selected tomato is determinate is $\frac{19}{30}$. 1

(ii) Mario buys ten tomatoes. Find the probability, correct to two decimal places, that no more than eight of these are determinate. 2

Hint: what is p for one tomato?

PP 10 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q9** 1 mark

The chance of being sunny on three consecutive days is 0.512. The probability that it is Sunny on exactly two days during a period of n days is 0.0512. What is the value of n? Assume the probability of being Sunny is the same every day.

A. 4

B. 5

C. 6

D. 7

Hint: what is p for one day?

LESSON 17B

Binomial distributions

Now that we have dealt with individual binomial probabilities, we can look at the whole distribution. The binomial distribution is a discrete distribution, and Chapter 13 of the Year 11 book was devoted to discrete distributions. As we have done before, we will draw its graph and work with its mean and variance. For $X \sim B(n, p)$, the mean is $\mu = np$ and the variance is $\sigma^2 = npq$. The distribution is symmetric when $p = \frac{1}{2}$, and skewed otherwise.

CONCEPT 1 · notation, and the Bernoulli distribution

NOTATION FOR BINOMIAL DISTRIBUTIONS

Some notation is convenient. The binomial distribution consisting of n independent trials, each with probability p of success, is denoted by $B(n, p)$ or $\mathbf{Bin}(n, p)$. In particular, a Bernoulli experiment with probability p is a special case of a binomial experiment, so is denoted by $B(1, p)$.

To say that X is a binomial random variable for the distribution $B(n, p)$, we can use the symbolic notation

$$X \sim B(n, p) \qquad \text{OR} \qquad X \sim \mathbf{Bin}(n, p).$$

In particular, $X \sim B(1, p)$ means that X is a Bernoulli random variable (there is only one stage) with probability p of success.

2 BINOMIAL AND BERNOULLI DISTRIBUTIONS

- The binomial distribution with n independent Bernoulli stages, each with probability p of success, is denoted by $B(n, p)$ or $\mathbf{Bin}(n, p)$.
- The Bernoulli distribution with probability p is therefore $B(1, p)$.
- $X \sim B(n, p)$ means that X is a random variable for the distribution $B(n, p)$.

THE MEAN AND VARIANCE OF A BERNOULLI DISTRIBUTION

A Bernoulli trial with probabilities p of 'success' and $q = 1 - p$ of 'failure' has mean p and variance pq,

$$\mu = p \qquad \text{and} \qquad \sigma^2 = pq,$$

so that its standard deviation is $\sigma = \sqrt{pq}$. These formulae are easy to prove because there are only two outcomes (see the worked example below).

TUTOR DEMO **Notation, and a Bernoulli variable**

a Write each random variable in the form $X \sim B(n, p)$.

(i) X is the number of heads when a coin is tossed 9 times.

(ii) X is the number of times a 5 or a 6 shows when a die is thrown 12 times.

(iii) X is 1 if a card drawn from a pack of 52 cards is a club, and 0 if it is not.

b $Y \sim B(1, 0.35)$. Use $\mu = \sum yP(Y = y)$ and $\sigma^2 = \sum y^2 P(Y = y) - \mu^2$ to find the mean, the variance and the standard deviation of Y, correct to three significant figures where necessary.

YOUR TURN **Same skill**

a Write each random variable in the form $X \sim B(n, p)$.

(i) X is the number of tails when a coin is tossed 15 times.

(ii) X is the number of ones when a die is thrown 10 times.

(iii) X is 1 if a die shows a six, and 0 if it does not.

b $Y \sim B(1, 0.8)$. Use $\mu = \sum yP(Y = y)$ and $\sigma^2 = \sum y^2 P(Y = y) - \mu^2$ to find the mean, the variance and the standard deviation of Y.

WORKED EXAMPLE **From the text: keep for homework**

Prove that a Bernoulli trial with probabilities p of 'success' and $q = 1 - p$ of 'failure' has mean $\mu = p$ and variance $\sigma^2 = pq$.

SOLUTION

We use the formula $\mu = \sum xP(X = x)$, where the sum is taken over the distribution, meaning for $x = 0$ and $x = 1$,

$$\begin{aligned}\text{so} \quad \mu &= 0 \times P(X = 0) + 1 \times P(X = 1) \\ &= 0 \times q + 1 \times p \\ &= p.\end{aligned}$$

Similarly, summing over the distribution,

$$\begin{aligned}\sigma^2 &= \sum x^2 P(X = x) - \mu^2 \\ &= (0^2 \times q + 1^2 \times p) - p^2 \\ &= p - p^2 \\ &= p(1 - p) \\ &= pq \quad (\text{either form is an appropriate answer}).\end{aligned}$$

CONCEPT 2 · the mean and variance of a binomial distribution

THE MEAN AND VARIANCE OF A BINOMIAL DISTRIBUTION

The mean and variance of the binomial random variable $X \sim B(n, p)$ with n trials and probabilities p and q of 'success' and 'failure' are

$$\mu = np \quad \text{and} \quad \sigma^2 = npq,$$

so that the standard deviation is $\sigma = \sqrt{npq}$.

These two results $\mu = np$ and $\sigma^2 = npq$ seem to follow immediately from the results $\mu = p$ and $\sigma^2 = pq$ for Bernoulli trials just by multiplying by n. And they do: it is true in general that if we have a number of independent random variables, then the mean of the sum is the sum of the means, and the variance of the sum is the sum of the variances.

Unfortunately, however, those two very general theorems are too difficult to prove, and instead we must prove the result for binomial distributions directly. The proofs below are not conceptually difficult, but they do require a sequence of computations. We begin with a lemma about binomial coefficients.

A LEMMA ABOUT BINOMIAL COEFFICIENTS

Lemma: Let n and x be whole numbers with $x \leq n$.

a If $x \geq 1$, then $\quad x \times {}^nC_x = n \times {}^{n-1}C_{x-1}$.

b If $x \geq 2$, then $\quad x(x-1) \times {}^nC_x = n(n-1) \times {}^{n-2}C_{x-2}$.

Proof:

a

$$\begin{aligned} x \times {}^nC_x &= x \times \frac{n!}{x! \times (n-x)!} \\ &= \frac{x \times n \times (n-1)!}{x \times (x-1)! \times (n-x)!} \\ &= n \times \frac{(n-1)!}{(x-1)! \times (n-x)!} \\ &= \text{RHS} \end{aligned}$$

b

$$\begin{aligned} & x(x-1) \times {}^nC_x \\ &= x(x-1) \times \frac{n!}{x! \times (n-x)!} \\ &= \frac{x(x-1) \times n(n-1) \times (n-2)!}{x(x-1) \times (x-2)! \times (n-x)!} \\ &= n(n-1) \times \frac{(n-2)!}{(x-2)! \times (n-x)!} \\ &= \text{RHS} \end{aligned}$$

PROVING THAT THE MEAN IS np

This proof requires binomial expansions, together with part **a** of the lemma above.

$$\begin{aligned} \mathrm{E}(X) &= \sum xP(X=x), \text{ summed over the distribution,} \\ &= 0 \times {}^nC_0 \times p^0q^n + 1 \times {}^nC_1 \times p^1q^{n-1} + 2 \times {}^nC_2 \times p^2q^{n-2} + \cdots \\ &\quad + n \times {}^nC_n \times p^nq^0. \end{aligned}$$

The first term is 0, then we apply part **a** of the lemma to the remaining terms,

$$\begin{aligned} \mathrm{E}(X) &= n \times {}^{n-1}C_0\, p^1q^{n-1} + n \times {}^{n-1}C_1\, p^2q^{n-2} + \cdots + n \times {}^{n-1}C_{n-1}\, p^nq^0 \\ &= np\left({}^{n-1}C_0\, p^0q^{n-1} + {}^{n-1}C_1\, p^1q^{n-2} + \cdots + {}^{n-1}C_{n-1}\, p^{n-1}q^0\right). \end{aligned}$$

The bit in brackets is the binomial expansion of $(p+q)^{n-1}$, where $p+q=1$, so

$$\begin{aligned} \mathrm{E}(X) &= np(p+q)^{n-1} \\ &= np \times 1 \\ &= np. \end{aligned}$$

PROVING THAT THE VARIANCE IS npq

This proof again requires binomial expansions. First, however, we need to develop yet another formula for the variance.

$$\begin{aligned} \mathrm{Var}(X) &= \mathrm{E}(X^2) - \big(\mathrm{E}(X)\big)^2 \\ &= \mathrm{E}\big(X(X-1)\big) + \mathrm{E}(X) - \big(\mathrm{E}(X)\big)^2 \\ &= \mathrm{E}\big(X(X-1)\big) + np - n^2p^2, \quad \text{because } \mathrm{E}(X) = np. \end{aligned}$$

We can find $\mathrm{E}\big(X(X-1)\big)$ using binomial expansions and part **b** of the lemma,

$$\begin{aligned}\mathrm{E}\big(X(X-1)\big) &= \sum x(x-1)P(X=x), \text{ summed over the distribution,}\\ &= 0\times(-1)\times {}^nC_0\times p^0q^n + 1\times 0\times {}^nC_1\times p^1q^{n-1}\\ &\quad +2\times 1\times {}^nC_2\times p^2q^{n-2}\\ &\quad +3\times 2\times {}^nC_3\times p^3q^{n-3}+\cdots+n(n-1)\times {}^nC_n\times p^nq^0.\end{aligned}$$

The first two terms are 0, then applying part **b** of the lemma,

$$\begin{aligned}&= n(n-1)\times {}^{n-2}C_0\times p^2q^{n-2} + n(n-1)\times {}^{n-2}C_1\times p^3q^{n-3}\\ &\quad +n(n-1)\times {}^{n-2}C_2\times p^4q^{n-4}+\cdots+n(n-1)\times {}^{n-2}C_{n-2}\times p^nq^0\\ &= n(n-1)p^2\big({}^{n-2}C_0\,p^0q^{n-2} + {}^{n-2}C_1\,p^1q^{n-3}+\cdots+{}^{n-2}C_{n-2}\,p^{n-2}q^0\big).\end{aligned}$$

The bit in brackets is the binomial expansion of $(p+q)^{n-2}$, where $p+q=1$, so

$$\begin{aligned}&= n(n-1)p^2(p+q)^{n-2}\\ &= n^2p^2-np^2.\end{aligned}$$

$$\begin{aligned}\text{Hence } \mathrm{Var}(X) &= \mathrm{E}\big(X(X-1)\big)+np-n^2p^2\\ &= n^2p^2-np^2+np-n^2p^2\\ &= np(1-p)\\ &= npq.\end{aligned}$$

3 MEAN AND VARIANCE OF A BINOMIAL DISTRIBUTION

- For a binomial random variable $X\sim B(n,p)$, where $q=1-p$,
  $\mu=np \quad \text{and} \quad \sigma^2=npq \quad \text{and} \quad \sigma=\sqrt{npq}.$
- In particular, for a Bernoulli random variable $X\sim B(1,p)$,
  $\mu=p \quad \text{and} \quad \sigma^2=pq \quad \text{and} \quad \sigma=\sqrt{pq}.$

NOTATION

The symbols $\mathrm{E}(X)$ and μ are interchangeable: μ is more concise, but the notation $\mathrm{E}(X)$ indicates that when we run the experiment, there is a sense in which we are 'expecting' to get $\mathrm{E}(X)$. Similarly, σ^2 and $\mathrm{Var}(X)$ are interchangeable.

TUTOR DEMO $n = 25, p = 0.2$

A binomial random variable has parameters $n = 25$ and $p = 0.2$.
a Find the mean, variance and standard deviation.
b What is the probability of getting the mean when the experiment is run? Give your answer correct to three decimal places.
c What is the probability that the result is within one standard deviation of the mean? Give your answer correct to three decimal places.
d Give an example that this distribution could model.

YOUR TURN $n = 18, p = \frac{1}{3}$

A binomial random variable has parameters $n = 18$ and $p = \frac{1}{3}$.
a Find the mean, variance and standard deviation.
b What is the probability of getting the mean when the experiment is run? Give your answer correct to three decimal places.
c What is the probability that the result is within one standard deviation of the mean? Give your answer correct to three decimal places.
d Give an example that this distribution could model.

WORKED EXAMPLE 6 **Keep for homework**

A binomial random variable has parameters $n = 20$ and $p = 0.1$.
a Find the mean, variance and standard deviation.
b What is the probability of getting the mean when the experiment is run?
c What is the probability that the result is within one standard deviation of the mean?
d Give an example that this distribution could model.

SOLUTION

a

$$\begin{aligned}\mu &= np \\ &= 20 \times 0.1 \\ &= 2\end{aligned} \qquad \begin{aligned}\sigma^2 &= npq \quad (\text{where } q = 1 - p = 0.9) \\ &= 20 \times 0.1 \times 0.9 \\ &= 1.8\end{aligned} \qquad \begin{aligned}\sigma &= \sqrt{1.8} \\ &\doteq 1.342\end{aligned}$$

b

$$\begin{aligned}P(X = 2) &= {}^{20}C_2\, p^2 q^{18} \\ &= {}^{20}C_2 \times (0.1)^2 \times (0.9)^{18} \\ &\doteq 0.285.\end{aligned}$$

c

$$\begin{aligned}P(\mu - \sigma \leq X \leq \mu + \sigma) &= P(0.658 \leq X \leq 3.342) \\ &= P(X = 1 \text{ or } X = 2 \text{ or } X = 3) \\ &= {}^{20}C_1\, p^1 q^{19} + {}^{20}C_2\, p^2 q^{18} + {}^{20}C_3\, p^3 q^{17} \\ &= 20 \times (0.1) \times (0.9)^{19} + 190 \times (0.1)^2 \times (0.9)^{18} \\ &\quad + 1140 \times (0.1)^3 \times (0.9)^{17} \\ &\doteq 0.745.\end{aligned}$$

d Choose a busy intersection with traffic lights on the way to work. On 20 mornings, when crossing at the lights, look at the number plate of the left-most front vehicle stopped at the lights, and record whether the last digit-character is a 7. In a small country town, it is just possible that the events may not be independent, but in a city, independence is virtually certain.

CONCEPT 3 · conditions on the mean and standard deviation

FINDING n OR p

The formulae in Box 3 can also be used to find n or p when a condition is placed on the mean or the standard deviation. A condition on σ is usually easier to use after squaring, because $\sigma^2 = npq$ has no square root. Both sides are positive, so squaring keeps the inequality sign. When n is fixed and the condition is on p, the result is a quadratic inequality in p (see Worked example 7).

FINDING n AND p FROM μ AND σ^2

If the mean μ and the variance σ^2 are both given, divide $\sigma^2 = npq$ by $\mu = np$:

$$q = \frac{\sigma^2}{\mu}, \qquad \text{then} \qquad p = 1 - q \qquad \text{and} \qquad n = \frac{\mu}{p}.$$

Check that n is a whole number.

TUTOR DEMO $n = 400, p = 0.1$

A binomial random variable has parameters $n = 400$ and $p = 0.1$.

a Find the mean and standard deviation.

b Keeping $p = 0.1$, what would n need to be increased to so that the standard deviation is less than 1% of the number n of trials?

c Keeping $n = 400$, what must p be decreased to so that the standard deviation is less than 3? Can p be increased to give a standard deviation less than 3?

d A different binomial random variable has mean 10 and variance 6. Find p and n.

YOUR TURN $n = 225, p = 0.2$

A binomial random variable has parameters $n = 225$ and $p = 0.2$.
a Find the mean and standard deviation.
b Keeping $p = 0.2$, what would n need to be increased to so that the standard deviation is less than 2% of the number n of trials?
c Keeping $n = 225$, what must p be decreased to so that the standard deviation is less than 3? Can p be increased to give a standard deviation less than 3?
d A different binomial random variable has mean 12 and variance 3. Find p and n.

WORKED EXAMPLE 7 **Keep for homework**

A binomial random variable has parameters $n = 100$ and $p = \frac{1}{5}$.
a Find the mean and standard deviation.
b Keeping $p = \frac{1}{5}$, what would n need to be increased to so that the standard deviation is less than 1% of the number n of trials?
c Keeping $n = 100$, what must p be decreased to so that the standard deviation is less than 1? Can p be increased to give a standard deviation less than 1?

SOLUTION

a

$$\begin{aligned}\mu &= np \\ &= 100 \times \tfrac{1}{5} \\ &= 20,\end{aligned} \qquad \begin{aligned}\sigma^2 &= npq \quad (\text{where } q = 1 - p = \tfrac{4}{5}) \\ &= 100 \times \tfrac{1}{5} \times \tfrac{4}{5} \\ &= 16,\end{aligned} \qquad \begin{aligned}\sigma &= \sqrt{16} \\ &= 4.\end{aligned}$$

b Put $\sigma < \dfrac{n}{100}$.

$$\text{Squaring,} \quad npq < \frac{n^2}{10\,000}$$

$$n^2 > 10\,000 \times \tfrac{4}{25} \times n, \quad \text{because } pq = \tfrac{1}{5} \times \tfrac{4}{5} = \tfrac{4}{25}.$$

$$\text{Hence} \quad n > 1600, \quad \text{because } n \text{ is positive.}$$

c Put $\sigma < 1$.

$$\begin{aligned}\text{Squaring,} \quad npq &< 1 \\ 100p(1-p) &< 1 \\ 100p^2 - 100p + 1 &> 0.\end{aligned}$$

The quadratic $100p^2 - 100p + 1 = 0$, with $a = 100$, $b = -100$ and $c = 1$, has axis of symmetry $p = \frac{1}{2}$ and discriminant $\Delta = 9600 = 40^2 \times 6$,

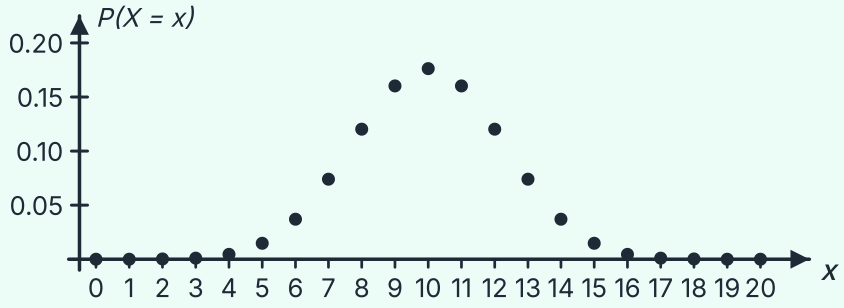
$$\begin{aligned}\text{so its roots are} \quad p &= \frac{100 - 40\sqrt{6}}{200} \quad \text{and} \quad p = \frac{100 + 40\sqrt{6}}{200} \\ &= \tfrac{1}{2} - \tfrac{1}{5}\sqrt{6} \qquad\qquad = \tfrac{1}{2} + \tfrac{1}{5}\sqrt{6} \\ &\doteq 0.0101 \qquad\qquad \doteq 0.9899,\end{aligned}$$

so p should be decreased to less than $\frac{1}{2} - \frac{1}{5}\sqrt{6}$ (roughly, less than 0.01), or increased to more than $\frac{1}{2} + \frac{1}{5}\sqrt{6}$ (roughly, greater than 0.99).

CONCEPT 4 · skewed and non-skewed binomial distributions

SYMMETRIC WHEN $p = q = \frac{1}{2}$

If $p = q = \frac{1}{2}$, such as when we toss a coin a number of times, the distribution is symmetric. For example, in Section 15E (Year 11) we graphed the distribution when a coin is tossed 20 times and the number of heads is recorded.

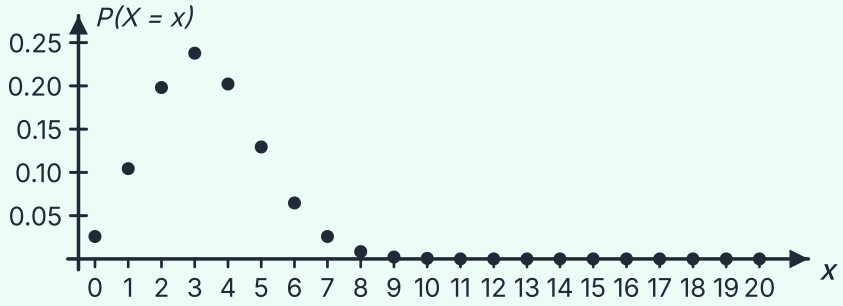


The mean is $\mu = np = 10$, and the distribution is symmetric about $x = 10$, with standard deviation $\sigma = \sqrt{npq} = \sqrt{5} \doteq 2.236$. This symmetry is easily seen from the symmetry of the binomial coefficient. For example, ${}^{20}C_7 = {}^{20}C_{13}$, so

$$P(X = 7) = {}^{20}C_7 \times \left(\tfrac{1}{2}\right)^7 \times \left(\tfrac{1}{2}\right)^{13} = {}^{20}C_{13} \times \left(\tfrac{1}{2}\right)^{13} \times \left(\tfrac{1}{2}\right)^7 = P(X = 13).$$

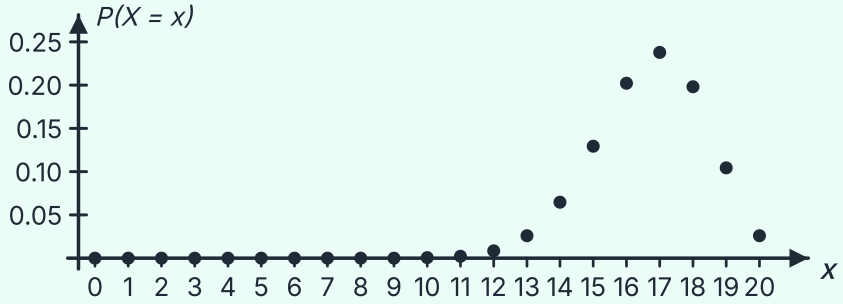
SKEWED TO THE RIGHT, AND SKEWED TO THE LEFT

When, however, we throw a die 20 times, recording 'success' when the result is 6, the distribution is decidedly skewed to the right (positive skewness). Here $p = \frac{1}{6}$ and $q = \frac{5}{6}$, with mean $\mu = 3\frac{1}{3} \doteq 3.333$ and standard deviation $\sigma = \frac{5}{3} \doteq 1.667$.



Remember that *skewed to the right* means that there is a *longer tail on the right-hand side*, and the peak is on the left-hand side.

On the other hand, when we throw the die 20 times but record 'success' as 'not getting 6', then the distribution is skewed to the left (negative skewness). Here $p = \frac{5}{6}$ and $q = \frac{1}{6}$, with mean $\mu = 16\frac{2}{3} \doteq 16.667$ and the same standard deviation $\sigma = \frac{5}{3} \doteq 1.667$.



Again, remember that *skewed to the left* means that there is a *longer tail on the left-hand side*, and the peak is on the right-hand side.

The graphs are reflections of each other in $x = 10$. For example,

$$P(7 \text{ sixes}) = {}^{20}C_7 \times \left(\tfrac{1}{6}\right)^7 \times \left(\tfrac{5}{6}\right)^{13} = {}^{20}C_{13} \times \left(\tfrac{5}{6}\right)^{13} \times \left(\tfrac{1}{6}\right)^7 = P(13 \text{ non-sixes}).$$

Question 11 in Exercise 17B shows how, for a fixed number of trials, the standard deviation decreases as the distribution becomes more skewed. Worked example 8 is a simple example.

4 SKEWED BINOMIAL DISTRIBUTIONS

Suppose that a binomial distribution consists of n independent Bernoulli trials, each with probability p of success and probability $q = 1 - p$ of failure.

- Reversing 'success' and 'failure' reverses the probabilities p and q, and reflects the graph in $x = \frac{1}{2}n$.
- If $p = q = \frac{1}{2}$, then the distribution is symmetric about $x = \frac{1}{2}n$.
- If $p < \frac{1}{2}$, the distribution is skewed to the right (positive skewness).
  If $p > \frac{1}{2}$, the distribution is skewed to the left (negative skewness).
- For distributions with the same number of trials, the standard deviation gets smaller as p gets closer to 0 or 1, that is, as the skewness becomes more pronounced.

TUTOR DEMO $p = \frac{1}{2}$ **and** $p = \frac{1}{5}$

Two binomial random variables X and Y both consist of 100 Bernoulli trials. For X, $p = q = \frac{1}{2}$, and for Y, $p = \frac{1}{5}$.
a Find the ratio of their standard deviations.
b Describe the shape of each distribution.
c Z counts the failures in the experiment for Y. Find the mean and the standard deviation of Z, and describe the shape of its distribution.

YOUR TURN $p = \frac{1}{2}$ and $p = \frac{1}{4}$

Two binomial random variables X and Y both consist of 48 Bernoulli trials. For X, $p = q = \frac{1}{2}$, and for Y, $p = \frac{1}{4}$.

a Find the ratio of their standard deviations.

b Describe the shape of each distribution.

c Z counts the failures in the experiment for Y. Find the mean and the standard deviation of Z, and describe the shape of its distribution.

WORKED EXAMPLE 8 **Keep for homework**

Two binomial random variables X and Y both consist of 100 Bernoulli trials. For X, $p = q = \frac{1}{2}$, and for Y, $p = \frac{1}{10}$. Find the ratio of their standard deviations.

SOLUTION

For X, $\sigma_X^2 = npq$
$= 100 \times \frac{1}{2} \times \frac{1}{2}$
$= 25,$
so $\sigma_X = 5.$

For Y, $\sigma_Y^2 = npq$
$= 100 \times \frac{1}{10} \times \frac{9}{10}$
$= 9,$
so $\sigma_Y = 3.$

Hence the ratio of the two standard deviations is $\sigma_X : \sigma_Y = 5 : 3$.

CONCEPT 5 · simulating the experiment and graphing the data

SIMULATIONS

The probabilities in a binomial distribution are estimates of what will happen when the experiment is run. Let us see what happens in simulations of two binomial experiments (Worked examples 9 and 10). The first experiment is symmetric, the second is skewed to the right.

Calculation and simulation in both worked examples should be done in a spreadsheet (the binomial function is currently `BINOM.DIST` in Excel). In this lesson the results of each simulation are given, so the rest can be done with a calculator.

- The relative frequency of each value is $f_r = \dfrac{f}{\text{number of runs}}$.
- The mean $\bar{x}$ and the standard deviation s of the data are found from the frequency table, as in Chapter 13 of the Year 11 book.
- The histogram has columns 1 unit wide, centred on the values of x. The polygon joins the midpoints of the tops of the columns.

TUTOR DEMO **Eight coins**

The experiment is, ‘toss 8 coins and count the number of heads’.
a Find the mean and standard deviation, and complete a table of the probability distribution (probabilities correct to three decimal places).
b The experiment was simulated 100 times, with these results. Complete the table of relative frequencies, and find the mean and standard deviation of the data.

x	0	1	2	3	4	5	6	7	8
f	0	4	10	22	28	20	11	4	1

c Draw the theoretical frequency histogram and polygon from part **a**. Then draw the frequency histogram and polygon of the simulation in part **b**.

YOUR TURN **Eight dice**

The experiment is, ‘roll 8 dice and count the number of dice showing a 5 or a 6’.
a Find the mean and standard deviation, and complete a table of the probability distribution (probabilities correct to three decimal places).
b The experiment was simulated 100 times, with these results. Complete the table of relative frequencies, and find the mean and standard deviation of the data.

x	0	1	2	3	4	5	6	7	8
f	5	15	29	26	16	7	2	0	0

c Draw the theoretical frequency histogram and polygon from part **a**. Then draw the frequency histogram and polygon of the simulation in part **b**.

WORKED EXAMPLE 9 **Keep for homework**

The experiment is, 'toss 10 coins and count the number of heads'.
a Find the mean and standard deviation, and complete a table of the probability distribution.
b Simulate the experiment 100 times, complete the table of relative frequencies, and find the mean and standard deviation of the data.
c Draw the theoretical frequency histograms and polygon from part **a**. Then draw the frequency histograms and polygon of the simulation in part **b**.

SOLUTION

a Here $p = q = \frac{1}{2}$ and $n = 10$, so

$$\mu = np = 5, \qquad \sigma^2 = npq = 2.5, \qquad \sigma = \sqrt{2.5} \doteq 1.58.$$

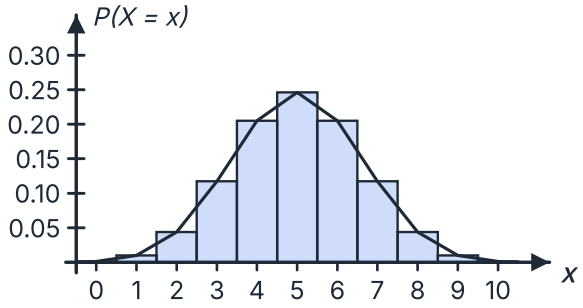
x	0	1	2	3	4	5	6	7	8	9	10
$P(X = x)$	0.001	0.010	0.044	0.117	0.205	0.246	0.205	0.117	0.044	0.010	0.001

b After running the experiment 100 times,

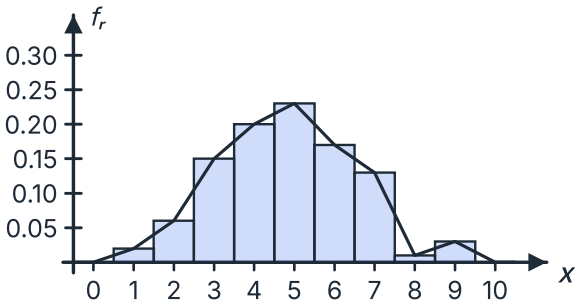
x	0	1	2	3	4	5	6	7	8	9	10
f	0	2	6	15	20	23	17	13	1	3	0
f_r	0	0.02	0.06	0.15	0.2	0.23	0.17	0.13	0.01	0.03	0

Calculation gives $\bar{x} = 4.82$ and $s \doteq 1.70$.

c



The theoretical distribution



The simulation

WORKED EXAMPLE 10 **Keep for homework**

The experiment is, 'roll 10 dice and count the number of sixes'.
a Find the mean and standard deviation, and complete a table of the probability distribution.
b Simulate the experiment 100 times, complete the table of relative frequencies, and find the mean and standard deviation of the data.
c Draw the theoretical frequency histograms and polygon from part **a**. Then draw the frequency histograms and polygon of the simulation in part **b**.

SOLUTION

a Here $p = \frac{1}{6}$ and $q = \frac{5}{6}$ and $n = 10$, so

$$\mu = np = 1.667, \qquad \sigma^2 = npq = 1.389, \qquad \sigma \doteq 1.179.$$

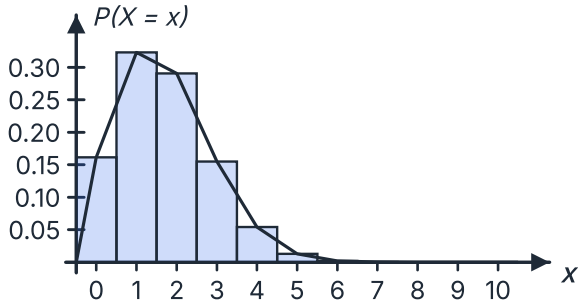
x	0	1	2	3	4	5	6	7	8	9	10
$P(X = x)$	0.162	0.323	0.291	0.155	0.054	0.013	0.002	0.0002	0.00002	0	0

b After running the experiment 100 times,

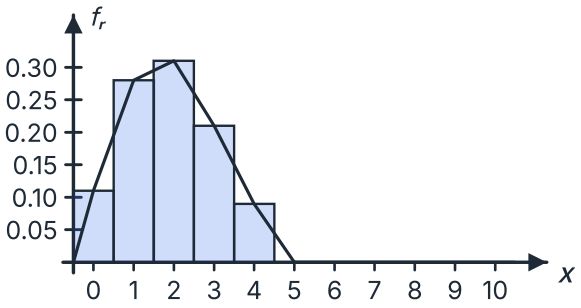
x	0	1	2	3	4	5	6	7	8	9	10
f	11	28	31	21	9	0	0	0	0	0	0
f_r	0.11	0.28	0.31	0.21	0.09	0.00	0.00	0.00	0.00	0.00	0.00

Calculation gives $\bar{x} = 1.89$ and $s \doteq 1.13$.

c



The theoretical distribution



The simulation

TEXTBOOK HOMEWORK · Exercise 17B

FOUNDATION

1 Identify which of the following experiments may be modelled using a binomial random variable. If so, identify the variable.

a It rains four days in a row. The number of rainy days is recorded.

b A die is thrown ten times, and after each throw it is recorded whether the result is less than five.

c Maddy is playing a simple game of cards. She turns up a card from the top of the pack. If it is an ace of spades she wins. Otherwise the card is returned to the pack, which is shuffled. The number of plays until she wins is recorded.

d The probability of a head on tossing an unfair coin is 0.4. A coin is tossed twenty times and the number of tails is recorded.

e Quality control testers have been given a random sample of 20 pens from a batch. It is known that 3% of the pens in the batch are defective. The testers record the number of faulty pens in the sample.

f A pupil records the time for his journey to school over a thirty-day period.

g At stage n an experimenter selects one number from 1–100 and records a success if the number matches n. This experiment is repeated 50 times (that is, up to $n = 50$) and the experimenter wishes to know how many successes occur.

2 In a simple experiment, 6 fair coins are tossed and the number of heads is recorded. A student wishes to compare the results with theoretical predictions.

a Copy and complete the table below using the formula $P(X = x) = {}^nC_x\, p^x q^{n-x}$ for binomial probability.

Number of heads	0	1	2	3	4	5	6	Total
Number of ways can occur								
Probability								

b Read off the mode (the most common result).

c Use this table to determine the expected value and variance of this discrete probability distribution.

d Compare your results to those obtained using the two formulae $\mathrm{E}(X) = np$ and $\mathrm{Var}(X) = npq$, where $q = 1 - p$.

e Explain, in your own words, why the result for the expected value is not a surprise.

3 For each situation, construct a table for the theoretical distribution associated with the given random variable and calculate the mode, mean and standard deviation.

a A coin is tossed five times and X is the number of heads that are face up.

b A die is thrown five times and X is the number of times 5 or 6 occurs.

c Five cards are drawn in turn with replacement from a standard pack of 52 cards and X is the number of court cards (jack, queen, king) that are drawn. Give your result correct to 3 decimal places.

4 At the School fête, few people have entered any of the 24 meat raffles, so to support the school, Larry buys 8 of the 20 tickets in each raffle.

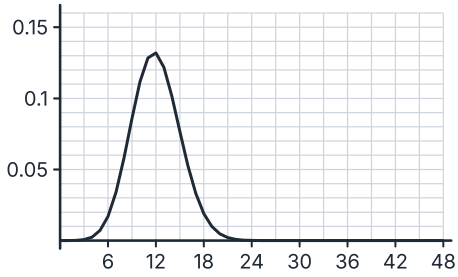
a Calculate the expected value and standard deviation for the random variable of the number of Larry's wins.

b If Larry wins 6 raffles, calculate how many standard deviations Larry's result is below the mean, as a measure of his poor luck.

5 **a** If a player throws 6 dice, what is the probability of getting at least two sixes? Let the random variable X record the number of sixes. What is the expected value and standard deviation for X?

b Repeat the question for:
(i) 12 dice, (ii) 24 dice.

6 The graph to the right shows the relative frequency polygon for the binomial distribution with $n = 48$ and $p = 0.25$.



a Is the distribution skewed to the left (negatively) or to the right (positively) or not at all?

b Find the mean and standard deviation for the distribution.

c What is the mode of the distribution? Estimate the probability of this most likely outcome.

d Copy the graph and shade the region no more than two standard deviations from the mean.

e Use the given sketch for $p = 0.25$ to assist in sketching the graph of the distribution for $p = 0.75$.

DEVELOPMENT

8 Ms Taylor sets her class a test of 48 multiple-choice questions, each with four options A–D. Let the random variable X be the number of questions a person gets correct.

a Calculate the expected value $\mathrm{E}(X)$. How can this value be understood in this context?

b What is the standard deviation σ of the random variable?

c Ms Taylor is annoyed to discover that Fayola, one of her students, claims to have got 24 just by guessing. How many standard deviations is Fayola's score above the mean?

Ms Taylor writes a new test, with 100 questions, each with five options A–E.

d What is the expected value and standard deviation of this new distribution with random variable Y?

e Fayola gets 40 questions right this time, and again she claims to have achieved this result just by guessing. If true, would this be more or less unusual than her previous result?

f Idette gets 75% in the first test and 60% in the second. Which is the more unusual of her results, if she is guessing?

9 A certain drug is found to be helpful for 70% of patients who take it. Two research teams are attempting to improve its effectiveness by adjusting the delivery system for the drug. Both conduct random trials to test the effectiveness of their changes. Team A runs a trial with 50 patients, of whom 45 show improvement using the drug. Team B runs a trial with 90 patients, of whom 74 show improvement. Use the mean and standard deviation of the two trials to decide which adjustment shows stronger evidence of an improvement.

10 It is known that 15% of a large constituency voted for the Working Together Party (WTP) at the last election. A poll of 100 people is taken and they are asked whether they voted WTP.

a What is the mean and standard deviation for this poll?

b What is the probability that in the sample, the number of WTP voters lies within half a standard deviation of the mean?

11 In this question, take the number n of stages of a binomial variable $X \sim B(n, p)$ to be fixed, and allow p to vary.

a Find σ^2 as a quadratic in p, and graph it.

b Explain the symmetry of the graph.

c Show that $\sigma^2 < np$ and $\sigma^2 < nq$.

d Show that the maximum value of σ^2 is $\frac{n}{4}$.

e Let n be fixed. Show that $\sigma \to 0$ as $p \to 0^+$ and as $p \to 1^-$.

ENRICHMENT

14 In experimental trials, it may be useful to ensure that there is at least one positive result. For example, the trial may be difficult or expensive to run, and at least one successful result may be required. Suppose that in a certain set of n independent Bernoulli trials, the probability of success at each stage is p.

d A certain journal refuses to publish experimental studies that do not yield a positive result, meaning a result agreeing with the experimental hypothesis. Explain why this is a dangerous practice.

15 If a Bernoulli trial occurs a fixed number n of times, and if the stages are independent, then the resulting distribution is a *binomial distribution*. If instead the experiment continues until a success is obtained and the number of trials is recorded, the resulting distribution is called a *geometric distribution* (and is also called a *discrete waiting-time distribution*).

Suppose that at each stage the probability of success is p and the probability of failure is $q = 1 - p$, and that X is the first trial producing a success. Thus the possible values of X are $1,\ 2,\ 3,\ \ldots$.

a For this geometric distribution:
(i) Show that $P(X = x) = pq^{x-1}$.
(ii) Show that $\mu = p + 2pq + 3pq^2 + \cdots$.
(iii) Show that $\mu - q\mu = p + pq + pq^2 + \cdots$, and hence calculate μ.

b During the day at a certain medical practice, the waiting room is constantly at capacity, with 20 patients waiting to be seen by a doctor. Every 5 minutes, a new patient will be chosen at random to see a doctor and a new patient will arrive at the practice. What is the mean waiting time to see a doctor?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **21 marks** Calculator allowed · show working · hints are below each question if you get stuck

PP 1 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q1** 1 mark

The random variable X represents the number of successes in 12 independent Bernoulli trials. The probability of success is $p = 0.7$ in each trial.
Which of the following statements correctly state the mean and variance of X?

A. The mean is 8.4 and the variance is 1.59
B. The mean is 8.4 and the variance is 2.52
C. The mean is 12 and the variance is 1.59
D. The mean is 12 and the variance is 2.52

Hint: which formulae in Box 3 give the mean and the variance of $B(n, p)$?

PP 2 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q6** 1 mark

A random variable X is defined by $P(X = k) = \binom{15}{k}(0.29)^k(0.71)^{15-k}$ for $k = 0, 1, 2, \ldots, 15$.
What is the mean of X?

A. 0.29
B. 0.71
C. 4.35
D. 10.65

Hint: $\binom{15}{k}$ means ${}^{15}C_k$. Which binomial distribution is this?

PP 3 **Blacktown Boys' High School, 2023, Year 12 Trial Examination, Mathematics Extension 1, Q5** 1 mark

The random variable X is such that $X \sim Bin(n, p)$. The mean value of X is 225 and variance of X is 144. What is the value of p?

A. 0.8
B. 0.64
C. 0.6
D. 0.36

Hint: what do you know about n, p and q?

PP 4 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)(i)(ii)** 2 marks

The probability that a certain drug cures a disease is 0.8. The drug is administered to 10 000 patients.
Let X be the binomial random variable representing the number of patients cured.

(i) Find $E(X)$. 1

(ii) Show that X has a standard deviation of 40. 1

Hint: what are n and p here?

PP 5 **Baulkham Hills High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q11(c)(ii)** 2 marks

Two fair six-sided dice are rolled together and the sum of the numbers on their upper faces is recorded. The result is recorded as either a 7 or not a 7.
In an earlier part of the question, it was found that the probability of a sum of 7 is $p = \frac{1}{6}$.

(ii) The two dice are rolled 30 times. Find the mean and variance of this binomial distribution. 2

Hint: which binomial distribution $B(n, p)$ is this?

PP 6 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(i)** **1 mark**

In a certain school, 23% of Year 12 students study Mathematics Extension 1.

(i) If X represents the number of students who study Mathematics Extension 1, describe the skewness of the binomial distribution for $P(X = x)$. 1
You must give reasons for your answer.

Hint: look at Box 4. What decides the direction of the skew?

PP 7 **Knox Grammar School, 2022, Higher School Certificate Trial Examination, Mathematics Extension 1, Q6** **1 mark**

The random variable X is distributed binomially such that $X \sim Bin(n, 0.2)$.
If the mean of X is double the size of the standard deviation of X, what is the size of n?

A. 2
B. 8
C. 12
D. 16

Hint: what are the mean and the standard deviation of X?

PP 8 **Blacktown Boys' High School, 2022, HSC Trial Examination, Mathematics Extension 1, Q13(d)** **2 marks**

A coin is tossed 30 times.
Let X be the number of heads. Find the probability that the number of heads is within one standard deviation of the mean using the binomial distribution. Round off your answer to 3 decimal places.

Hint: which binomial distribution is this?

PP 9 **Sydney Technical High School, 2023, Extension 1 Mathematics, Trial HSC, Q12(a)** 6 marks

Given that $X \sim Bin(n, p)$, $\mu = 12$ and $\sigma^2 = 4$

(i) Show that $p = \dfrac{2}{3}$ 2

(ii) Hence find n 1

(iii) By considering a six-sided die or otherwise, describe a possible real-world scenario that could be represented by this binomial distribution. 1

(iv) Find $P(X = 4)$ 2

Hint: for (iii), which events on one throw of a die have probability $\frac{2}{3}$?

PP 10 **Moriah College, 2023, Trial Higher School Certificate Examination, Mathematics Extension 1, Q12(a)** 4 marks

A box contains 3 blue marbles, 5 green marbles and k yellow marbles.

A marble is chosen at random and then placed back into the box. This is repeated four times.

(i) What is the probability that the first marble chosen is blue? 1

(ii) Let X be the number of blue marbles chosen. 3

Find the smallest value of k for which $\text{Var}(X) < 0.8$

Hint: what are n and p here?

LESSON 17C

Normal approximations to a binomial

Calculating binomial probabilities exactly becomes dreadful when the numbers are large. Fortunately, binomial distributions can be approximated very well by the normal distribution, and the larger the numbers are, the better the approximation. This lesson approximates $B(n, p)$ by $N(np, npq)$, using the continuity correction for small values of n, and the rule of thumb $np > 5$ and $nq > 5$. Use the standard normal table at the end of the Revision section, as in Chapter 16.

CONCEPT 1 · the normal approximation to a binomial

WHY APPROXIMATE?

A pollster asks a sample of 10 000 people whether they intend to vote for the Working Together Party in the next election. The WTP won $\frac{1}{3}$ of the vote in the last election, and using the assumption that this will be the case in the next election, he wants to know the probability that his survey will give him a result within $[3300, 3400]$.

Using binomial probability, the calculation is

$$\begin{aligned} P(3300 \leq X \leq 3400) = {}& {}^{10\,000}C_{3300}\left(\tfrac{1}{3}\right)^{3300}\left(\tfrac{2}{3}\right)^{6700} \\ & + {}^{10\,000}C_{3301}\left(\tfrac{1}{3}\right)^{3301}\left(\tfrac{2}{3}\right)^{6699} + \cdots \\ & + {}^{10\,000}C_{3399}\left(\tfrac{1}{3}\right)^{3399}\left(\tfrac{2}{3}\right)^{6601} \\ & + {}^{10\,000}C_{3400}\left(\tfrac{1}{3}\right)^{3400}\left(\tfrac{2}{3}\right)^{6600}. \end{aligned}$$

This is a dreadful calculation because of all the huge factorials and massive sizes of the numbers. Even with a computer, there are serious problems about controlling all the errors in the calculation. And if the calculation involves the whole population of Australia (over 25 000 000), the situation is even more difficult.

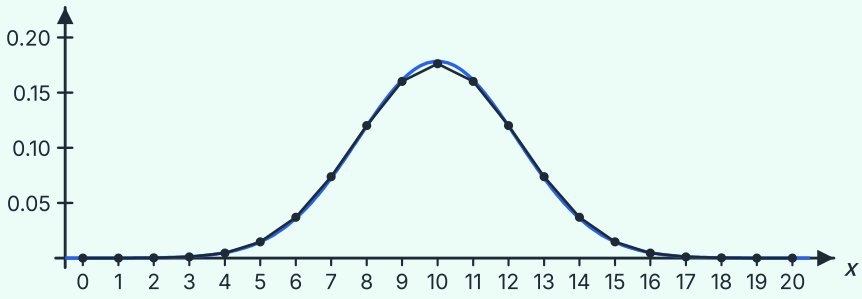
Fortunately, binomial distributions can be approximated very well by the normal distribution, and the larger the numbers are, the better the approximation. This section is about approximating a binomial distribution by a normal distribution.

AN EXAMPLE OF APPROXIMATIONS

In Section 16E, we took the binomial distribution of the number of heads in 20 coin tosses, where $n = 20$ and $p = q = \frac{1}{2}$, so that

$$\mu = np = 20 \times \tfrac{1}{2} = 10 \qquad \text{and} \qquad \sigma^2 = npq = 20 \times \tfrac{1}{2} \times \tfrac{1}{2} = 5,$$

and $\sigma = \sqrt{5}$. We graphed the frequency polygon, then we superposed a normal curve with the same mean $\mu = 10$ and the same standard deviation $\sigma = \sqrt{5}$. The result looks reasonably good,



Thus the normal PDF approximates the frequency polygon of the binomial, and taking cumulative frequencies, it follows that the normal CDF approximates the ogive of the binomial. We cannot in this course give theoretical arguments why such approximations work: we can only look at the pictures and confirm the results. Then we can use these results in problems.

For example, the standard deviation is $\sqrt{5} \doteq 2.236$. Suppose that we want to find the probability that when we toss 20 coins, the number of heads will be within one standard deviation of the mean. These numbers are small enough for us to calculate it two ways, as in Worked example 12.

NOTATION FOR NORMAL DISTRIBUTIONS

The notation $N(\mu, \sigma^2)$ is used for the normal distribution with mean μ and variance σ^2. Thus in the diagram above, we are claiming that the binomial distribution $B\left(20, \frac{1}{2}\right)$ is approximated by the normal distribution $N(10, 5)$. In general, $B(n, p)$ is approximated by $N(np, npq)$, where $q = 1 - p$.

TREATING X AS IF IT WERE CONTINUOUS (WORKED EXAMPLE 11)

To approximate the binomial by the normal distribution $N(10, 5)$, we treat the discrete random variable X *as if it were a continuous normal variable*. Because we are using a continuous density function, we need to follow the boundaries of the histograms and find the probability, not that $X \geq 13$, but that $X \geq 12.5$, which is the left-hand boundary of the first histogram. This adjustment from 13 to 12.5 is called the *continuity correction*: it is the correction needed when a discrete distribution is treated as if it were continuous.

Then find the z-score $z = \dfrac{x - \mu}{\sigma}$ and use the standard normal table at the end of the Revision section, with z correct to two decimal places, or a calculator.

TUTOR DEMO **A coin tossed 16 times**

A coin is tossed 16 times, and X is the number of heads. Find the probability that the number of heads is greater than 10, correct to three decimal places:
a using a binomial distribution,
b using the normal approximation (use the standard normal table, with z correct to two decimal places).

YOUR TURN **A coin tossed 12 times**

A coin is tossed 12 times, and X is the number of heads. Find the probability that the number of heads is greater than 8, correct to three decimal places:
a using a binomial distribution,
b using the normal approximation (use the standard normal table, with z correct to two decimal places).

WORKED EXAMPLE 11 **Keep for homework**

A coin is tossed 20 times, and X is the number of heads. Find the probability that the number of heads is greater than 12:
a using a binomial distribution,
b using the normal approximation.

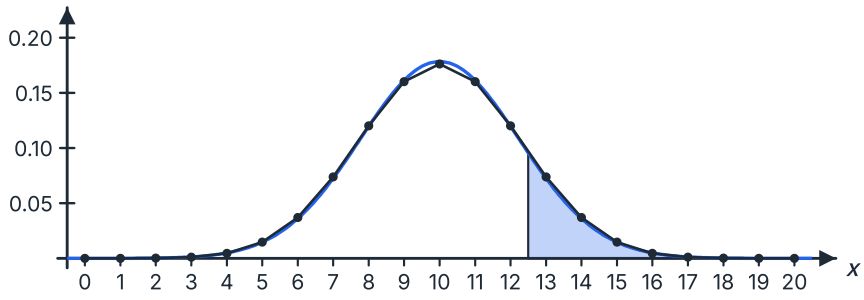
SOLUTION

a

$$\begin{aligned} P(\text{at least 13 heads}) &= {}^{20}C_{13}\left(\tfrac{1}{2}\right)^{20} + {}^{20}C_{14}\left(\tfrac{1}{2}\right)^{20} + \cdots + {}^{20}C_{20}\left(\tfrac{1}{2}\right)^{20} \\ &= \left(\tfrac{1}{2}\right)^{20}\left({}^{20}C_{13} + {}^{20}C_{14} + \cdots + {}^{20}C_{20}\right) \\ &= \left(\tfrac{1}{2}\right)^{20} \times 137\,980 \\ &\doteq 0.132. \end{aligned}$$

b To approximate the binomial by the normal distribution $N(10, 5)$, we treat the discrete random variable X *as if it were a continuous normal variable*. Because we are using a continuous density function, we need to follow the boundaries of the histograms and find the probability, not that $X \geq 13$, but that $X \geq 12.5$, which is the left-hand boundary of the first histogram.

$$P(\text{at least 13 heads}) \doteq P(X \geq 12.5).$$



This adjustment from 13 to 12.5 is called the *continuity correction*: it is the correction needed when a discrete distribution is treated as if it were continuous.

$$\text{Using } z\text{-scores}, \quad \begin{aligned} z &= \frac{x - \mu}{\sigma} \\ &= \frac{12.5 - 10}{\sqrt{5}} \\ &\doteq 1.118, \end{aligned}$$

$$\begin{aligned} \text{so } P(\text{at least 13 heads}) &\doteq P(Z > 1.118) \\ &\doteq 1 - \Phi(1.118) \\ &\doteq 1 - 0.868 \\ &\doteq 0.132. \end{aligned}$$

CONCEPT 2 · continuity corrections

CONTINUITY CORRECTIONS

We are approximating a discrete binomial variable by a continuous normal variable. We thus integrate from halfway between the values of the discrete distribution. In Worked example 11, that means from 12.5.

In Worked example 12, we are adding the five values at $x = 8,\ 9,\ 10,\ 11$ and 12, so we integrate over the interval $[7.5, 12.5]$.

5 NORMAL APPROXIMATION TO A BINOMIAL DISTRIBUTION

- A binomial distribution can be approximated by the normal distribution with the same mean and standard deviation.
- Thus the binomial distribution $B(n, p)$ or $\mathrm{Bin}(n, p)$ can be approximated by the normal distribution $N(np, npq)$, where $q = 1 - p$.
- $N(\mu, \sigma^2)$ means the normal distribution with mean μ and variance σ^2.
- When approximating, we treat the discrete binomial variable X as if it were a continuous normal variable.
- For small values of n, apply the *continuity correction*. This means integrating between half-intervals, corresponding to the boundaries of the cumulative frequency histogram. For example, to approximate $P(X = 8,\ 9,\ 10,\ 11 \text{ or } 12)$, we treat X as a continuous normal variable and find
  $P(7.5 \leq X \leq 12.5)$.

WRITING THE CONTINUITY CORRECTION

First write the event with whole numbers and $\leq$ or $\geq$, then move half a unit outwards, so that every column of the event is included in full. For a whole number a:

Event	$X = a$	$X \leq a$	$X < a$, that is, $X \leq a - 1$	$X \geq a$	$X > a$, that is, $X \geq a + 1$
Use	$a - 0.5 \leq X \leq a + 0.5$	$X \leq a + 0.5$	$X \leq a - 0.5$	$X \geq a - 0.5$	$X \geq a + 0.5$

TUTOR DEMO **Continuity corrections, and 16 coins**

a $X \sim B(30, 0.4)$. Write each probability as the probability for a continuous normal variable, using the continuity correction.

(i) $P(X = 12)$ (ii) $P(X \leq 10)$ (iii) $P(X < 10)$

(iv) $P(X \geq 15)$ (v) $P(X > 15)$ (vi) $P(8 \leq X < 14)$

b A coin is tossed 16 times, and X is the number of heads. Find the probability that the number of heads is within one standard deviation of the mean, correct to three decimal places, using a binomial distribution, and then using the normal approximation (use the standard normal table). Comment on the results.

YOUR TURN **Continuity corrections, and 12 coins**

a $X \sim B(40, 0.25)$. Write each probability as the probability for a continuous normal variable, using the continuity correction.

(i) $P(X = 10)$ (ii) $P(X \leq 7)$ (iii) $P(X < 12)$

(iv) $P(X \geq 13)$ (v) $P(X > 6)$ (vi) $P(5 < X \leq 9)$

b A coin is tossed 12 times, and X is the number of heads. Find the probability that the number of heads is within one standard deviation of the mean, correct to three decimal places, using a binomial distribution, and then using the normal approximation (use the standard normal table). Comment on the results.

WORKED EXAMPLE 12 **Keep for homework**

A coin is tossed 20 times, and X is the number of heads. Find the probability that the number of heads is within one standard deviation of the mean:

a using a binomial distribution,

b using the normal approximation.

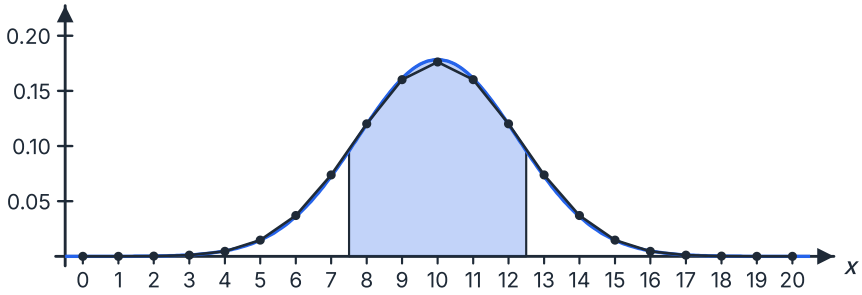
Comment on the results of the calculations.

SOLUTION

a

$$\begin{aligned}P(X \text{ is within one standard deviation of the mean}) &= P(7.764 \leq X \leq 12.236) \\ &= P(X = 8,\ 9,\ 10,\ 11 \text{ or } 12) \\ &= {}^{20}C_8\left(\tfrac{1}{2}\right)^{20} + {}^{20}C_9\left(\tfrac{1}{2}\right)^{20} + {}^{20}C_{10}\left(\tfrac{1}{2}\right)^{20} + {}^{20}C_{11}\left(\tfrac{1}{2}\right)^{20} + {}^{20}C_{12}\left(\tfrac{1}{2}\right)^{20} \\ &= \frac{{}^{20}C_8 + {}^{20}C_9 + {}^{20}C_{10} + {}^{20}C_{11} + {}^{20}C_{12}}{2^{20}} \\ &= \frac{772\,616}{1024 \times 1024} \\ &\doteq 0.737.\end{aligned}$$

b We are approximating the binomial distribution by the normal distribution $N(10, 5)$, so we treat the discrete random variable X as if it were a continuous normal variable, and calculate $P(7.5 \leq X \leq 12.5)$.



Using z-scores, the two limits of integration are

$$\begin{aligned}z &= \frac{x - \mu}{\sigma} \\ &= \frac{12.5 - 10}{\sqrt{5}} \\ &= 1.118,\end{aligned} \qquad \begin{aligned}z &= \frac{x - \mu}{\sigma} \\ &= \frac{7.5 - 10}{\sqrt{5}} \\ &= -1.118,\end{aligned}$$

$$\begin{aligned}\text{so } P(X \text{ is within one standard deviation of the mean}) &\doteq P(-1.118 \leq Z \leq 1.118) \\ &\doteq 0.736.\end{aligned}$$

The results in parts **a** and **b** are in excellent agreement.

CONCEPT 3 · the continuity correction for large numbers

THE CONTINUITY CORRECTION FOR LARGE NUMBERS

In the example in the introduction to this lesson (the pollster), the numbers are large enough so that the continuity correction is negligible, as Worked example 13 shows.

TUTOR DEMO $n = 10\,000, p = 0.2$

A binomial random variable X has $n = 10\,000$ and $p = 0.2$. Use the normal approximation and the standard normal table to find $P(1950 \leq X \leq 2060)$, correct to four decimal places:

a ignoring the continuity correction,

b using the continuity correction.

Then comment on the results of the calculations.

YOUR TURN $n = 4800, p = 0.25$

A binomial random variable X has $n = 4800$ and $p = 0.25$. Use the normal approximation and the standard normal table to find $P(1150 \leq X \leq 1260)$, correct to four decimal places:

a ignoring the continuity correction,

b using the continuity correction.

Then comment on the results of the calculations.

WORKED EXAMPLE 13 **Keep for homework**

Compute $P(3300 \leq X \leq 3400)$ in the example in the introduction:
a ignoring the continuity correction,
b using the continuity correction.
Then comment on the results of the calculations.

SOLUTION

Here $n = 10\,000$ and $p = \frac{1}{3}$, so $\mu = 3333\frac{1}{3}$ and $\sigma^2 = 2222\frac{2}{9}$ and $\sigma \doteq 47.1405$.
We will treat the discrete random variable X as if it were a continuous normal variable with the same mean $3333\frac{1}{3}$ and standard deviation 47.1405.

a Without the continuity correction, we are finding $P(3300 \leq X \leq 3400)$.
The z scores are -0.7071 and 1.4142,

$$\text{so} \quad P(3300 \leq X \leq 3400) \doteq P(-0.7071 \leq Z \leq 1.4142)$$
$$\doteq 0.6816.$$

b With the continuity correction, we are finding $P(3299.5 \leq X \leq 3400.5)$.
The z scores are -0.7177 and 1.4248,

$$\text{so} \quad P(3299.5 \leq X \leq 3400.5) \doteq P(-0.7177 \leq Z \leq 1.4248)$$
$$\doteq 0.6864.$$

No pollster would ever need the third decimal place of the probabilities, so there is no problem whatsoever ignoring the continuity correction.

CONCEPT 4 · using the normal approximation to calculate a binomial coefficient

APPROXIMATING A BINOMIAL COEFFICIENT

One of the consequences of the normal approximation to the binomial is that for large parameters, an individual binomial coefficient can be approximated. These calculations demonstrate very well how vital the continuity correction can be: without it, the probability that a continuous variable takes one exact value is zero.

When $p = \frac{1}{2}$, ${}^nC_x \times \left(\frac{1}{2}\right)^n = P(X = x)$, so
${}^nC_x = 2^n \times P(X = x) \doteq 2^n \times P(x - 0.5 \leq X \leq x + 0.5)$.

TUTOR DEMO $^{36}C_{20}$

Approximate $^{36}C_{20}$ using the normal approximation to the binomial distribution $B\left(36, \frac{1}{2}\right)$ and the standard normal table. Give your answer correct to four significant figures. Check how good the approximation is by finding $^{36}C_{20}$ on your calculator.

YOUR TURN $^{100}C_{55}$

Approximate $^{100}C_{55}$ using the normal approximation to the binomial distribution $B\left(100, \frac{1}{2}\right)$ and the standard normal table. Give your answer correct to four significant figures. Check how good the approximation is by finding $^{100}C_{55}$ on your calculator.

WORKED EXAMPLE 14 **Keep for homework**

Approximate ${}^{100}C_{53}$ using the normal approximation to the binomial distribution $B\left(100, \frac{1}{2}\right)$. Check how good the approximation is.

SOLUTION

We know that ${}^{100}C_{53} \times \left(\frac{1}{2}\right)^{100} = P(X = 53)$.
We treat the discrete binomial variable X as if it were a continuous normal variable,

$${}^{100}C_{53} \times \left(\frac{1}{2}\right)^{100} = P(52.5 \leq X \leq 53.5).$$

The binomial distribution has mean $\mu = np = 50$ and SD $\sigma = \sqrt{npq} = \sqrt{25} = 5$.
So also does the normal approximation, and the limits have z-scores 0.5 and 0.7.

$$\begin{aligned} \text{Hence} \quad {}^{100}C_{53} \times \left(\frac{1}{2}\right)^{100} &\doteq P(0.5 \leq Z \leq 0.7) \\ &\doteq 0.066\,574 \\ {}^{100}C_{53} &\doteq 2^{100} \times 0.066\,574 \\ &\doteq 8.439 \times 10^{28}. \end{aligned}$$

This compares with ${}^{100}C_{53} \doteq 8.441 \times 10^{28}$ (all according to Excel).

CONCEPT 5 · when is a survey binomial, and how good is the approximation?

SAMPLING WITH REPLACEMENT: WHEN IS A SURVEY A BINOMIAL EXPERIMENT?

Suppose that I choose ten cards in succession *without replacement* from a pack of 52 cards and count the number of aces. This is not a binomial experiment because the probabilities of getting an ace change as each card is removed from the pack. If *I replace the card each time*, however, then I have a binomial experiment.

More generally, a survey that questions n people chosen at random from N people without replacement is not a binomial experiment. A survey that chooses each person independently of all previous choices, however, thus allowing the same person possibly to be questioned more than once, is a binomial experiment because the stages are identical and independent.

Now suppose that the pool of N people from which the sample is chosen is very large. This allows us to ignore these distinctions, because whether replacement is used or not, the probabilities will be virtually the same in all stages. This is certainly the case with a sample chosen from all Australians, but in a college of say 3000 people and a sample that is small relative to 3000, the effect is also almost negligible.

HOW GOOD AN APPROXIMATION IS THE NORMAL?

This depends on how much accuracy is needed. The diagrams below show that n should be larger for a skewed distribution, meaning that p is near 0 or 1, than when the distribution

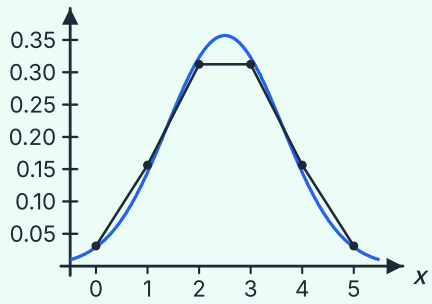
is reasonably symmetric, meaning that p is near 0.5. A common rule of thumb therefore is to require

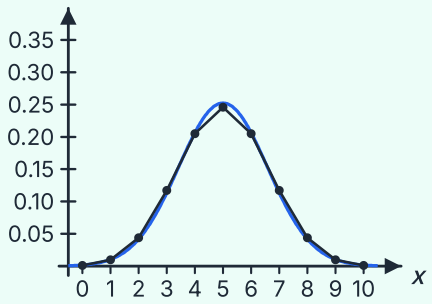
$$np > 5 \quad \text{and} \quad nq > 5,$$

but numbers other than 5 are often stated, and there are other tests.

When $p = \frac{1}{2}$, we have already drawn the approximation when $n = 20$. Here are the frequency polygon and the normal approximation, still for $p = \frac{1}{2}$, when $n = 5$ and $n = 10$.

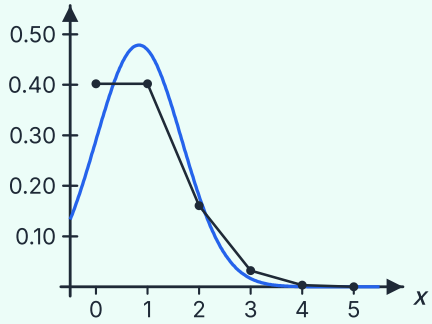
a

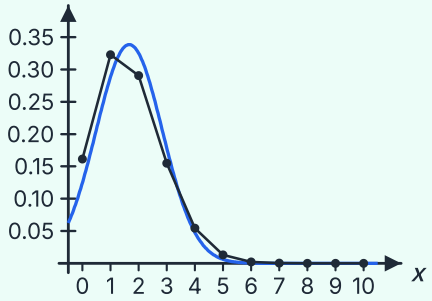


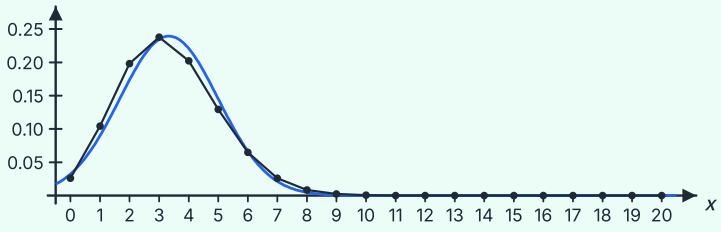


When $p = \frac{1}{6}$, which arises when throwing a die and recording if it is a six, here are the graphs for $n = 5$, $n = 10$ and $n = 20$.

b







6 A RULE OF THUMB FOR USING THE NORMAL APPROXIMATION

- For a fixed number n of Bernoulli trials, the normal approximation to a binomial distribution is less satisfactory when the distribution is skewed.
- For a fixed probability p, the normal approximation to a binomial distribution is more satisfactory when the number of trials is large.
- A common rule of thumb (one amongst many) for using the normal approximation to a binomial distribution is to require

$$np > 5 \quad \text{and} \quad nq > 5.$$

THE NORMAL APPROXIMATION TO A BINOMIAL DISTRIBUTION IS A SPECIAL CASE

We remarked at the start of Section 16G of the textbook, in the last chapter, that the normal approximation to a binomial distribution was one of the first examples of a very general theorem in statistics called the *central limit theorem*. This would be a good occasion to read again the short account of the central limit theorem given there.

TUTOR DEMO **Binomial or not? Is the approximation reasonable?**

a Is each experiment binomial? If not, is it very nearly binomial?

(i) Twenty cards are drawn in succession without replacement from a pack of 52 cards, and the number of hearts is counted.

(ii) Twenty cards are drawn in succession from a pack of 52 cards, each card being replaced before the next draw, and the number of hearts is counted.

(iii) A survey questions 50 people chosen at random from a city of 2 million people, and counts the number who own a dog.

b Use the rule of thumb in Box 6 to decide whether the normal approximation to each binomial distribution is reasonable.

(i) $B(40, 0.1)$ (ii) $B(60, 0.25)$ (iii) $B(200, 0.98)$

c According to the rule of thumb, how big does the sample need to be if $p = 0.04$?

YOUR TURN **Same skill**

a Is each experiment binomial? If not, is it very nearly binomial?

(i) Five marbles are drawn in succession without replacement from a bag with 6 red and 4 blue marbles, and the number of red marbles is counted.

(ii) A die is thrown 30 times, and the number of sixes is counted.

(iii) A survey questions 40 students chosen at random from a school of 1500 students, and counts the number who walk to school.

b Use the rule of thumb in Box 6 to decide whether the normal approximation to each binomial distribution is reasonable.

(i) $B(25, 0.3)$ (ii) $B(100, 0.03)$ (iii) $B(30, 0.9)$

c According to the rule of thumb, how big does the sample need to be if $p = 0.95$?

WORKED EXAMPLE **Testing the rule of thumb: keep for homework**

Test the rule of thumb $np > 5$ and $nq > 5$ on the six graphs in this lesson: $p = \frac{1}{2}$ with $n = 5$, 10 and 20 (Concept 1), and $p = \frac{1}{6}$ with $n = 5$, 10 and 20. Compare the results with the pictures.

SOLUTION

p	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$
n	5	10	20	5	10	20
np	2.5	5	10	0.83	1.67	3.33
nq	2.5	5	10	4.17	8.33	16.67
Rule of thumb	fails	fails (just)	holds	fails	fails	fails

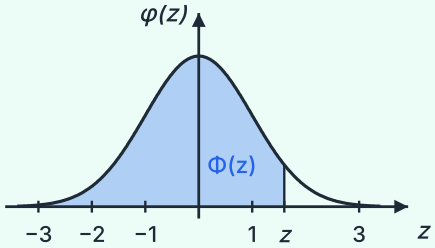
For $p = \frac{1}{2}$, the rule holds only when $n = 20$, and that graph fits very well. When $n = 10$, the fit already looks good, so the rule of thumb is a cautious one. For $p = \frac{1}{6}$, the distributions are skewed to the right and the rule fails even for $n = 20$, where $np \doteq 3.33$: the pictures show the poorer fit, especially near $x = 0$. By the rule of thumb, $p = \frac{1}{6}$ needs $n > 30$.

TEXTBOOK HOMEWORK · Exercise 17C

THE STANDARD NORMAL PROBABILITY DISTRIBUTION

The shaded area in the graph to the right represents a value of the *cumulative standard normal distribution function*

$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^{z} \phi(t)\, dt.$$



The standard normal table at the end of the Revision section gives values of the probabilities $P(Z \leq z) = \Phi(z)$, allowing two decimal places for z. After that, use interpolation. For example,

$$P(Z \leq 1.627) = \Phi(1.627) = \int_{-\infty}^{1.627} \phi(z)\, dz$$
$$\doteq 0.9474 + 0.7\,(0.9484 - 0.9474) \doteq 0.9481.$$

Note: You may find in this exercise and the next that your answers differ slightly from the text, depending whether or not you interpolate, and whether you use the supplied tables or alternatives such as statistical calculators and spreadsheets that provide more accurate values.

FOUNDATION

1 A certain binomial distribution for the random variable X consists of 20 independent trials, each with probability of success $p = 0.3$.

a Write this binomial distribution in symbolic form.

b Calculate the probability of obtaining 9, 10 or 11 successes.

c Confirm that $np > 5$ and $nq > 5$, suggesting that a normal approximation to the binomial could be used to estimate this probability.

d Calculate the normal approximation by determining $P(8.5 \leq X \leq 11.5)$, treating X as approximately $N(\mu, \sigma^2)$, and using the standard normal table at the end of the Revision section.

e Find the percentage error in the normal approximation. Does the approximation seem fairly accurate for this value of n and p?

2 Repeat the steps of Question 1 for these cases.

a $n = 50$, $p = 0.5$, find $P(18 \leq X \leq 20)$,

b $n = 20$, $p = 0.4$, find $P(8 \leq X \leq 9)$,

c $n = 30$, $p = 0.3$, find $P(5 \leq X \leq 7)$,

d $n = 40$, $p = 0.2$, find $P(9 \leq X \leq 12)$,

e $n = 22$, $p = 0.6$, find $P(13 \leq X \leq 15)$,

f $n = 80$, $p = 0.1$, find $P(10 \leq X \leq 13)$,

g $n = 500$, $p = 0.25$, find $P(100 \leq X \leq 103)$,

h $n = 200$, $p = 0.9$, find $P(170 \leq X \leq 172)$.

3 A barrel contains 600 pink and 400 blue counters. At each stage of an experiment, the barrel is stirred well, then a counter is removed, its colour is noted, and it is returned to the barrel. This experiment is repeated 20 times.

a Explain why each stage of the experiment is a Bernoulli trial.

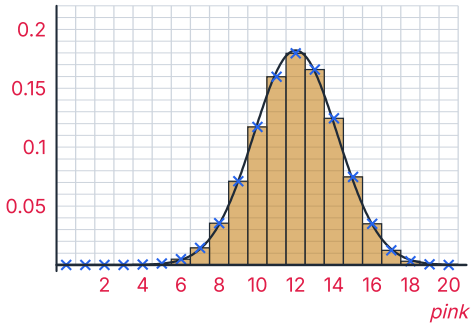
b Explain why the full experiment is binomial.

c Is it necessary to return the counter after each draw?

d Write down the probability of drawing a pink counter, and find the mean and standard deviation for this binomial distribution.

e Find the probability of drawing exactly 14 pink counters, correct to 3 decimal places.

f A student constructs a histogram for this distribution and overlays the normal distribution with the same mean and standard deviation. He notes the strong agreement, and to test this he uses his standard normal tables to calculate the area corresponding to drawing 14 pink counters.



(i) Explain why the required area for the normal distribution with random variable X is $P(13.5 < X < 14.5)$.

(ii) Hence find this area. Is it in strong agreement with your answer to part **e**?

5 A commonly used rule of thumb states that the normal approximation to a binomial distribution will be reasonable if $np > 5$ and $n(1-p) > 5$. This means that if p is further away from 0.5, the sample size needs to be bigger to get a reasonable approximation. According to this rule, how big does the sample need to be if:

a $p = 0.5$, **b** $p = 0.25$, **c** $p = 0.125$, **d** $p = 0.01$,
e $p = 0.75$, **f** $p = 0.875$, **g** $p = 0.9$, **h** $p = 0.55$?

6 According to some estimates, eight per cent of males in the world are colour-blind. A representative random sample includes 854 people, and a statistician wishes to determine the probability that between 7% and 9% of the people in the sample are colour-blind.

a Give a reason why the researcher might want to use a normal approximation to the binomial to calculate this probability.

b Comment on the assumption that the sample is representative and random.

c Calculate the required probability, using a normal approximation without a continuity correction.

d As a measure of the accuracy of ignoring any continuity correction, calculate the probability $P(76 < X < 76.5)$ associated with the boundary of 76 successes.

DEVELOPMENT

7 A horticulturalist is attempting to cross two species of flowers to strengthen certain characteristics. One of the plants has red flowers and the other's flowers are white, but the horticulturalist wishes to retain the strong red colour in the offspring. According to Mendel's theory of inheritance, there is a 25% chance that the flowers of the offspring will be red. The horticulturalist crosses 15 pairs of parent plants and notes the colour of their offspring plant. Find the probability that of the offspring:

a none will have red flowers,

b there will be at least one with red flowers,

c at least twenty per cent will have red flowers (use a normal approximation here).

8 Suppose that it is known that 45% of eighteen-year-olds in a particular city do not have a driver's licence. If a random sample of 20 eighteen-year-olds is taken in the city, what is the probability that more than half of them will not have a driver's licence?

9 Long-term studies show that 60% of the residents and visitors to Nashville Tennessee prefer Country music to Western. The local council provides Country music for those eating their lunch in the park to listen to. How confident can they be that in a group of thirty, more than twenty of them prefer Country? Comment on the assumption of independence in this question.

ENRICHMENT

12 Berry punnets distributed by a certain grower are exported in large batches. The receiver selects 10 punnets at random from each batch and tests whether any must be rejected because of rotten berries. The fraction p of punnets that are not of acceptable standard in the entire batch is unknown. The receiver needs a reasonable method of determining the quality of the batch without testing the entire batch, because the berries are destroyed in the process.

a One possible method is to reject the entire batch if any punnets in the sample are rejected.
(i) Suppose that $p = 0.05$, and determine the probability that the batch will be rejected.
(ii) Similarly determine and draw up a table showing the probability of rejection for $p = 0$, 0.1, 0.2, 0.3, 0.4. Include the case $p = 0.05$ in your table.
(iii) Plot a graph of p (on the horizontal axis) and the probability of rejecting the batch (on the vertical axis). This is called the *operating characteristic curve* (OC).

b Repeat part **a** if samples of 15 punnets are taken, but the batch is only rejected if two or more punnets are rejected. Include both plots on the same axes.

c Comment on whether part **a** or part **b** is the better method of ensuring the standard of the whole batch.

PAST-PAPER HOMEWORK · mini exam

Suggested time: 25 minutes **18 marks**

Calculator allowed · use the standard normal table at the end of the Revision section · show working · hints are below each question if you get stuck

Most of these questions do not say whether to use a continuity correction. The worked solutions follow each school's own solution, and say which method is used.

PP 1 **The King's School, 2021, Year 12 Alternative Higher School Certificate Trial Examination, Mathematics Extension 1, Q12(c)(iii)** 1 mark

The probability that a certain drug cures a disease is 0.8. The drug is administered to 10 000 patients.
Let X be the binomial random variable representing the number of patients cured.
In earlier parts of the question, it was found that $E(X) = 8000$ and that X has a standard deviation of 40.

(iii) Use a normal approximation to find the probability that X is less than 8040. 1

Hint: which normal distribution approximates X?

PP 2 **James Ruse Agricultural High School, 2020, Year 12 Trial HSC Examination, Mathematics Extension 1, Q12(b)** 2 marks

It is known that 80% of patients with a certain disease can be cured with a certain drug. What is the probability that amongst 150 patients with the disease, at most 37 of them cannot be cured with the drug? You must justify the use of the normal approximation.

Hint: what is the distribution of the number of patients who cannot be cured?

PP 3 **Ascham School, 2024, Year 12 Trial Examination, Mathematics Extension 1, Q11(e)** 3 marks

The percentage of left-handed people in a population is 10%. If one hundred people are selected at random from the population what is the probability that less than 5% are left-handed? [You may refer to the table at the end of the Revision section.]

Hint: which binomial distribution is this?

PP 4 **Killara High School, 2020, Mathematics Extension 1 Trial HSC Examination, Q12(e)** 3 marks

A restaurant knows from past experience that 27% of customers will order a dessert after the main course. For next week, the restaurant has taken 288 customer bookings.
Determine the probability that less than 100 will order dessert.
A normal distribution table is included [at the end of the Revision section].

Hint: what are n and p here?

PP 5 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(c)(ii)** 2 marks

You may use the information [in the standard normal table at the end of the Revision section] to answer this question.

A binomial variable, X, is known to have a probability of success, p, of close to 0.81, but the exact value is unknown.

In an earlier part of the question, a mathematician planned to perform n trials to find a more accurate estimate of p.

(ii) The mathematician repeats the experiment 2 million times, recording 1 615 580 successes. What is the probability that in 10 000 trials the number of successes would be greater than 8 100, correct as a percentage to 1 decimal place? 2

Hint: what do the 2 million trials tell you?

PP 6 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q13(d)** 3 marks

n students have been enrolled into the Higher Mathematics 1A course at university. It is seen that in every maths lecture, 5% of the students will be absent.

The probability that no more than 20 students miss the lecture is 0.90658.

Calculate the number of the students n, enrolled in the course.

(A Standard Normal Distribution table is [at the end of the Revision section].

Hint: what is the distribution of the number of students absent from a lecture?

PP 7 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q13(b)(ii)** 4 marks

In a large pool of people, it is known that 10% of them take less than 12 minutes to complete a particular test.

(ii) A random sample of n people is taken, the probability that fewer than 9 of these n people will take less than 12 minutes to complete the test is 0.3446, to 4 decimal places. Use normal approximation to solve for n, to the nearest integer. 4

Hint: what is the distribution of the number in the sample who take less than 12 minutes?

LESSON 17D

Sample proportions

So far, we have recorded the result of a binomial experiment as the number of successes. This lesson records instead the proportion of the Bernoulli trials that were successes, the *sample proportion* $\hat{p} = \dfrac{X}{n}$. Its mean is p and its variance is $\dfrac{pq}{n}$, so it estimates p, and for large n it is approximated by the normal distribution $N\left(p, \dfrac{pq}{n}\right)$. Use the standard normal table at the end of the Revision section.

CONCEPT 1 · the sample proportion and its mean and variance

SAMPLE PROPORTIONS

So far, we have recorded the result of a binomial experiment as the number of successes. Often, however, it is more interesting to record instead the *proportion of the Bernoulli trials that were successes*: this is called the *sample proportion*.

Sample proportions have many purposes, but one important purpose is to estimate the probability p of each Bernoulli trial in the binomial experiment. For example, before every election, surveys estimate the proportion of Australians voting for the Working Together Party. If a survey of 2000 finds 700 WTP voters (the number of successes), then the sample proportion is 0.35 (the proportion of successes), so the survey has estimated the WTP vote to be 35% of the population.

POPULATION PROPORTIONS

The probability p that a voter chosen at random will vote for the Working Together Party is a *population proportion*,

$$p = \frac{\text{number of Australians voting WTP}}{\text{number of Australian voters}}.$$

Many probabilities arise in this way, as we have often seen. Sample proportions apply to all binomial situations, but may be intuitively easier to understand if we think of the probability p arising as a population proportion, as in this voting example.

THE SAMPLE PROPORTION $\hat{p}$

Let $X \sim B(n, p)$ be a binomial variable consisting of n independent Bernoulli trials, each with probability p of success. The *sample proportion* $\hat{p}$ is the proportion of successes when the experiment is run. Thus *the sample proportion is a new random variable denoted by* $\hat{p}$ *and defined by*

$$\hat{p} = \frac{X}{n}.$$

For example, I survey 20 voters, and 7 say that they will vote WTP. This is a binomial experiment with 20 independent Bernoulli stages, and random variable X with 21 possible values $0,\ 1,\ 2,\ 3,\ \ldots,\ 20$. I can record my result as

$$X = 7 \qquad \text{or as} \qquad \hat{p} = \frac{7}{20}.$$

The binomial random variable X has 21 possible values $0,\ 1,\ 2,\ 3,\ \ldots,\ 20$, and the new random variable $\hat{p}$ has 21 possible values $0 = \frac{0}{20},\ \frac{1}{20},\ \frac{2}{20},\ \ldots,\ 1 = \frac{20}{20}$.

THE SAMPLE PROPORTION DISTRIBUTION: MEAN AND VARIANCE

A sample proportion random variable $\hat{p}$ is not a binomial random variable X because its values are fractions between 0 and 1 rather than whole numbers from 0 to n. But it is closely related to the corresponding binomial variable: the probability of each value of $\hat{p}$ is the same as the probability of the corresponding value of X,

$$P\left(\hat{p} = \frac{x}{n}\right) = P(X = x) = {}^{n}C_{x}\, p^{x} q^{n-x},$$

so the various graphs of a sample proportion distribution are just the graphs of the corresponding binomial distribution stretched horizontally by a factor of $\dfrac{1}{n}$.

The mean and variance of $\hat{p}$ come directly from the mean and variance of X,

$$\begin{aligned} E(\hat{p}) &= E\left(\frac{X}{n}\right) \\ &= \frac{E(X)}{n} \\ &= \frac{np}{n} \\ &= p, \end{aligned} \qquad \begin{aligned} \operatorname{Var}(\hat{p}) &= \operatorname{Var}\left(\frac{X}{n}\right) \\ &= \frac{\operatorname{Var}(X)}{n^2} \\ &= \frac{npq}{n^2} \\ &= \frac{pq}{n}, \end{aligned} \qquad \begin{aligned} \text{SD} &= \sqrt{\frac{npq}{n^2}} \\ &= \frac{\sqrt{npq}}{n}. \end{aligned}$$

The first formula $E(\hat{p}) = p$ proves what we said in the introduction to this lesson: running a binomial experiment and recording the sample proportion $\hat{p}$ gives an estimate of the Bernoulli probability p.

7 THE SAMPLE PROPORTION $\hat{p}$

Suppose that a binomial experiment with random variable X consists of n independent Bernoulli trials, each with probability p.

- The *sample proportion* is the random variable $\hat{p} = \dfrac{X}{n}$.
- The values of $\hat{p}$ are the $n+1$ values $0,\ 1,\ 2,\ \ldots,\ n$ of X divided by n. Thus the values of $\hat{p}$ are $0 = \frac{0}{n},\ \frac{1}{n},\ \frac{2}{n},\ \ldots,\ 1 = \frac{n}{n}$.
- The probability of each value of $\hat{p}$ equals the probability of the corresponding value of X,
  $$P\left(\hat{p} = \frac{x}{n}\right) = P(X = x) = {}^nC_x\, p^x q^{n-x}.$$
- The mean, variance and standard deviation of $\hat{p}$ are
  $$E(\hat{p}) = p, \qquad \text{Var}(\hat{p}) = \frac{pq}{n}, \qquad \text{standard deviation} = \frac{\sqrt{npq}}{n}.$$
- The sample proportion $\hat{p}$ gives an estimate of the probability p.

Here is a simple example comparing a binomial distribution with the corresponding sample proportion distribution (Worked example 15).

TUTOR DEMO **A spinner spun three times**

A spinner has three equal sectors, one red and two blue. It is spun three times. Let X be the number of reds, and $\hat{p}$ the corresponding sample proportion random variable.
a Tabulate the probabilities of X, find the mean and variance, and draw the frequency polygon.
b Tabulate the probabilities of $\hat{p}$, find the mean and variance, and draw the frequency polygon.

YOUR TURN **A coin tossed four times**

A coin is tossed four times. Let X be the number of heads, and $\hat{p}$ the corresponding sample proportion random variable.

a Tabulate the probabilities of X, find the mean and variance, and draw the frequency polygon.

b Tabulate the probabilities of $\hat{p}$, find the mean and variance, and draw the frequency polygon.

WORKED EXAMPLE 15 **Keep for homework**

A coin is tossed six times. Let X be the binomial variable and $\hat{p}$ the corresponding sample proportion random variable.

a Tabulate the probabilities of X, find the mean and variance, and draw the frequency polygon.

b Tabulate the probabilities of $\hat{p}$, find the mean and variance, and draw the frequency polygon.

SOLUTION

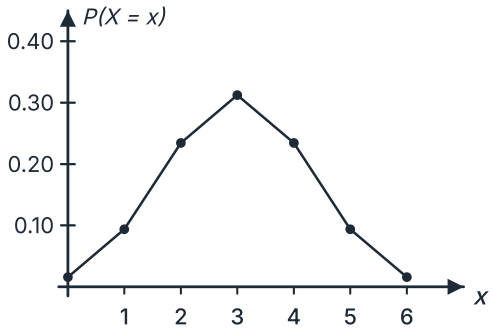
a Using the formula $P(x \text{ heads}) = {}^6C_x\left(\frac{1}{2}\right)^6$,

x	0	1	2	3	4	5	6
prob	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$

For the binomial variable X,

$E(X) = np = 3$ and

$\text{Var}(X) = npq = 1\frac{1}{2}$.



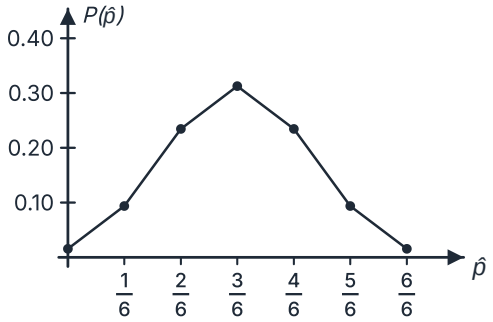
b The sample proportion is $\hat{p} = \dfrac{X}{n}$, and the corresponding probabilities are the same, so

$\hat{p}$	0	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1
prob	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$

For the sample proportion $\hat{p}$,

$E(\hat{p}) = p = \frac{1}{2}$ and

$\text{Var}(\hat{p}) = \dfrac{pq}{n} = \frac{1}{24}$.



CONCEPT 2 · using the sample proportion to estimate p

WHY ALL THIS MACHINERY OF SAMPLE PROPORTIONS?

Marketers and public opinion pollsters are very keen to find sample proportions for the Australian population. But surveys are expensive, and pollsters need to know how effective they can be with the small samples that they can afford or have time to carry out.

These things are also crucial for health research and public policy. Estimating a quantity such as the monthly unemployment statistics, or the number of Australians who require disability assistance services, is dependent on sample survey methods.

Working with the sample proportion not only provides an immediate estimate of p, but it also provides a straightforward approach to assessing how accurate the estimate is likely to be.

For example, Worked example 16 calculates the probability that a pollster will get within 5% and 10% of the correct result when using a sample of 20 voters to estimate the proportion voting for the WTP.

8 USING THE SAMPLE PROPORTION

The sample proportion $\hat{p}$ can often be used to estimate the probability of obtaining an acceptable estimate to the probability p of a Bernoulli experiment.

TUTOR DEMO **12 trials, $p = \frac{1}{4}$**

A binomial experiment with 12 Bernoulli trials is being used to estimate the probability p of success in the Bernoulli experiment. If the actual value of p is $\frac{1}{4}$, find the probability, correct to three decimal places, that the experiment will yield an estimate within:
a 0.05 of the actual value,
b 0.1 of the actual value.

YOUR TURN **15 trials, $p = \frac{2}{3}$**

A binomial experiment with 15 Bernoulli trials is being used to estimate the probability p of success in the Bernoulli experiment. If the actual value of p is $\frac{2}{3}$, find the probability, correct to three decimal places, that the experiment will yield an estimate within:
a 0.05 of the actual value,
b 0.1 of the actual value.

WORKED EXAMPLE 16 **Keep for homework**

A binomial experiment with 20 Bernoulli trials is being used to estimate the probability p of success in the Bernoulli experiment. If the actual value of p is $\frac{1}{3}$, find the probability that the experiment will yield an estimate within:
a 0.05 of the actual value,
b 0.1 of the actual value.

SOLUTION

Notice first that the estimate cannot be exactly $p = \frac{1}{3} \doteq 0.333$ because the size of the sample is 20, which is not a multiple of 3.

a To obtain an estimate within 0.05 of the actual value $p \doteq 0.333$, the sample proportion would need to be $\hat{p} = \frac{6}{20} = 0.30$ or $\hat{p} = \frac{7}{20} = 0.35$.

$$P\left(\hat{p} = \tfrac{6}{20} \text{ or } \hat{p} = \tfrac{7}{20}\right) = {}^{20}C_6\left(\tfrac{1}{3}\right)^6\left(\tfrac{2}{3}\right)^{14} + {}^{20}C_7\left(\tfrac{1}{3}\right)^7\left(\tfrac{2}{3}\right)^{13}$$
$$= 0.36425\ldots \quad \textbf{(save this value in the memory).}$$

Hence the probability that the estimate is within 0.05 of p is about 0.364.

b To obtain an estimate within 0.1 of the actual value $p \doteq 0.333$, the sample proportion would need to be $\hat{p} = \frac{5}{20} = 0.25$ or $\frac{6}{20}$ or $\frac{7}{20}$ or $\frac{8}{20} = 0.40$.

$$P\left(\hat{p} = \tfrac{5}{20} \text{ or } \hat{p} = \tfrac{8}{20}\right) = {}^{20}C_5\left(\tfrac{1}{3}\right)^5\left(\tfrac{2}{3}\right)^{15} + {}^{20}C_8\left(\tfrac{1}{3}\right)^8\left(\tfrac{2}{3}\right)^{12}$$
$$= 0.29368\ldots,$$

Adding the answer to part **a**, the probability that the estimate is within 0.1 of the actual value is about 0.658.

CONCEPT 3 · the normal approximation to the sample proportion

USING THE NORMAL APPROXIMATION TO THE SAMPLE PROPORTION

We discussed in Lesson 17C how to use normal approximations to a binomial distribution to obtain reasonable approximations to the binomial distribution far more quickly. The probabilities of the sample proportions are just the probabilities of the corresponding values of the binomial variable, so the same approximation methods can be used here. We cannot prove in this course that the normal distribution approximates $\hat{p}$: it is another example of the central limit theorem discussed at the start of Section 16G of the textbook, in the last chapter.

The normal approximation to the sample proportion thus has the same mean p as the sample proportion distribution, and the same variance $\frac{pq}{n}$, so we can approximate it using the normal distribution $N\left(p, \frac{pq}{n}\right)$.

As an example, let us increase the numbers in the previous worked example back up to realistic levels, and use the normal approximation to obtain the result. Political surveys before elections attempt to reduce the margin of error in their estimates to 1%–2%, that is, they attempt to estimate p correct to within 0.01–0.02.

WORKING WITH $\hat{p}$ IN PROBLEMS

- Write the event in terms of $\hat{p}$. For example, 'more than 22% of the 500 people' is $\hat{p} > 0.22$.
- Use the mean $\mu = p$ and the standard deviation $\sigma = \sqrt{\frac{pq}{n}}$, where p is the assumed population proportion.
- Standardise with $z = \frac{\hat{p} - p}{\sigma}$, then use the standard normal table, with z correct to two decimal places, or a calculator.
- The rule of thumb $np > 5$ and $nq > 5$ from Lesson 17C still tells us when the normal approximation is reasonable.
- The continuity correction is seldom used with $\hat{p}$, because the numbers are usually large. If it is used, the gap between the values of $\hat{p}$ is $\frac{1}{n}$, so move by half a gap, $\frac{1}{2n}$ (Worked example 19).

TUTOR DEMO **A survey of 1000 voters**

A survey of 1000 voters is being used to estimate the probability p that a random voter will vote for a particular party. Assuming that the actual value of p is 0.4, use the normal approximation to the sample proportion distribution and the standard normal table to find, correct to four decimal places, the probability that the survey will yield an estimate:
a within 0.02 of the actual value,
b within 0.03 of the actual value,
c greater than 0.42.

YOUR TURN **A survey of 1500 voters**

A survey of 1500 voters is being used to estimate the probability p that a random voter will vote for a particular party. Assuming that the actual value of p is 0.3, use the normal approximation to the sample proportion distribution and the standard normal table to find, correct to four decimal places, the probability that the survey will yield an estimate:
a within 0.01 of the actual value,
b within 0.02 of the actual value,
c less than 0.28.

WORKED EXAMPLE 17 **Keep for homework**

A survey of 2000 Australian voters is being used to estimate the probability p that a random voter will vote for the Working Together Party. Assuming that the actual value of p is $\frac{1}{3}$, use the normal approximation to the sample proportion distribution to find the probability that the experiment will yield an estimate:
a within 0.01 of the actual value,
b within 0.02 of the actual value.

SOLUTION

The numbers in the sample are large enough for us to ignore continuity corrections.
The normal approximation to $\hat{p}$ has

$$\text{mean} = \tfrac{1}{3} \doteq 0.3333 \qquad \text{and} \qquad \text{variance} = \frac{pq}{n} = \frac{2}{9} \times \frac{1}{2000} = \frac{1}{9000},$$

so taking the square root, the standard deviation is about 0.01054.

a We need to find $P(0.3233 \leq \hat{p} \leq 0.3433)$, so we find z-scores.

$$\text{For } 0.3233, \quad z = \frac{-0.01}{0.01054} \doteq -0.9487. \qquad \text{For } 0.3433, \quad z = \frac{0.01}{0.01054} \doteq 0.9487.$$

Hence the probability that the estimate is within 0.01 of the actual value is about

$$P(-0.9487 \leq Z \leq 0.9487) \doteq 0.66.$$

b We need to find $P(0.3133 \leq \hat{p} \leq 0.3533)$. The z-scores are now:

$$\text{For } 0.3133, \quad z = \frac{-0.02}{0.01054} \doteq -1.8974. \qquad \text{For } 0.3533, \quad z = \frac{0.02}{0.01054} \doteq 1.8974.$$

Hence the probability that the estimate is within 0.02 of the actual value is about

$$P(-1.8974 \leq Z \leq 1.8974) \doteq 0.94.$$

WORKED EXAMPLE 18 **Keep for homework**

Election surveys usually claim that their margin of error is about 2%, and the last worked example has addressed this. What are some other issues that pollsters need to take into account?

SOLUTION

- Can I be sure that the sample is unbiased?
- Are people answering honestly?
- Are opinions changing as the election approaches?
- What is to be done with people who refuse to answer or 'don't know'?
- Were there language or cultural problems that interfered with the interview?

CONCEPT 4 · the data, the sample proportion, and the normal approximation

THE DATA, THE SAMPLE PROPORTION, AND THE NORMAL APPROXIMATION

The situation has now become quite complicated because we have:

1. values of the sample proportion $\hat{p}$ obtained from surveys,
2. the sample proportion distribution, if we assume a value for p,
3. the normal approximation to the sample proportion distribution, also if we assume a value for p.

We conclude this chapter by drawing one more picture, using the dataset that we gained by simulation of a binomial experiment in Worked example 9 of Lesson 17B. We are interested in the distribution of the sample proportion $\hat{p}$, that is, the distribution of the estimate of the probability, and we will use the cumulative distributions in our comparisons.

Notice that the density function of the normal approximation to $\hat{p}$ is obtained from the density function of the approximation to X by stretching horizontally by a factor of $\frac{1}{n}$, and then stretching vertically by a factor of n so that the area under the curve remains 1. Therefore we can't superpose the graph of that normal density function on the graph of the frequency polygon because the heights don't match. Instead, we will compare the cumulative graphs to see the agreement.

TUTOR DEMO **Eight coins, from Lesson 17B**

In the tutor demo of Concept 5 in Lesson 17B, the experiment 'toss 8 coins and count the number of heads' was simulated 100 times, with these results.

x	0	1	2	3	4	5	6	7	8
f	0	4	10	22	28	20	11	4	1

Now imagine that each run gives an estimate of the probability of tossing heads, a value of $\hat{p}$, which has the 9 values $0, \frac{1}{8}, \frac{2}{8}, \ldots, 1$.

a From the data, produce a table showing the relative frequency and the cumulative relative frequency of each value of $\hat{p}$, and find the mean and standard deviation.

b Use binomial probability to calculate the cumulative probabilities of the values of $\hat{p}$, correct to three decimal places, and find the mean and standard deviation.

c Use the normal approximation to the sample proportion distribution, with the continuity correction, to calculate the cumulative probabilities of the values of $\hat{p}$. Give each z and $\Phi(z)$ correct to three decimal places.

d On one graph, draw the cumulative relative frequency polygon of the data, the cumulative probability polygon of the distribution, and the CDF of the normal approximation.

YOUR TURN **Eight dice, from Lesson 17B**

In the your-turn of Concept 5 in Lesson 17B, the experiment 'roll 8 dice and count the number of dice showing a 5 or a 6' was simulated 100 times, with these results.

x	0	1	2	3	4	5	6	7	8
f	5	15	29	26	16	7	2	0	0

Now imagine that each run gives an estimate of the probability of a 5 or a 6, a value of $\hat{p}$, which has the 9 values $0, \frac{1}{8}, \frac{2}{8}, \ldots, 1$.

a From the data, produce a table showing the relative frequency and the cumulative relative frequency of each value of $\hat{p}$, and find the mean and standard deviation.

b Use binomial probability to calculate the cumulative probabilities of the values of $\hat{p}$, correct to three decimal places, and find the mean and standard deviation.

c Use the normal approximation to the sample proportion distribution, with the continuity correction, to calculate the cumulative probabilities of the values of $\hat{p}$. Give each z and $\Phi(z)$ correct to three decimal places.

d On one graph, draw the cumulative relative frequency polygon of the data, the cumulative probability polygon of the distribution, and the CDF of the normal approximation.

WORKED EXAMPLE 19 **Keep for homework**

In Worked example 9 of Lesson 17B, we ran 100 simulations of the binomial experiment of tossing 10 coins and recording the number of heads. Now imagine that the purpose of each simulation was to produce an estimate of the probability of tossing heads, that is, a value of the random variable $\hat{p}$. Note that this random variable $\hat{p}$ has the 11 values $0 = \frac{0}{10}, \frac{1}{10}, \frac{2}{10}, \ldots, 1 = \frac{10}{10}$.

a From the data, produce a table showing the cumulative relative frequency of obtaining each value of $\hat{p}$, and find the mean and standard deviation.

b Use binomial probability to calculate the cumulative probabilities of obtaining the values of $\hat{p}$, and find the mean and standard deviation.

c Use the normal approximation to the sample proportion distribution to calculate the cumulative probabilities of values of $\hat{p}$, and find the mean and standard deviation. Apply the continuity correction, because $n = 10$ is small.

d On one graph, draw the cumulative relative frequency polygon of the data, the cumulative probability polygon of the distribution, and the CDF of the normal approximation.

SOLUTION

The calculations in parts **a** and **b** were done in Worked example 9, and we only need to divide the values through by 10.

a After running the experiment 100 times, each relative frequency is the estimated probability of obtaining each estimate of $\hat{p}$.

$\hat{p}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
f_r	0.00	0.02	0.06	0.15	0.2	0.23	0.17	0.13	0.01	0.03	0.00
cf_r	0.00	0.02	0.08	0.23	0.43	0.66	0.83	0.96	0.97	1.00	1.00

Calculation gives $\bar{x} = 0.482$ and $s \doteq 0.170$.

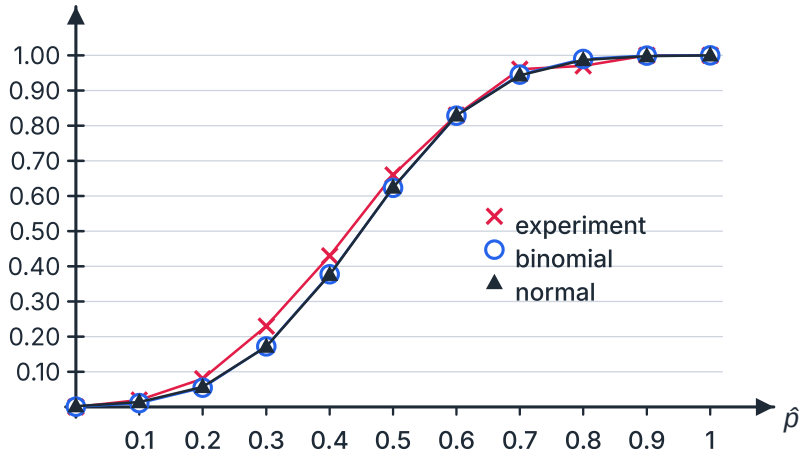
b From previous calculations, $\mu = 0.5$ and $\sigma = \frac{1}{10}\sqrt{2.5} \doteq 0.158$.

$\hat{p}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
$P(\hat{p})$	0.001	0.010	0.044	0.117	0.205	0.246	0.205	0.117	0.044	0.010	0.001
Cmlve	0.001	0.011	0.055	0.172	0.377	0.623	0.828	0.945	0.989	0.999	1.000

c The mean and standard deviation of the normal approximation are the same as for the sample proportion distribution. We need z-scores first, and we apply the continuity correction with 0.1 as the gaps in the values for $\hat{p}$. For example, for $\hat{p} = 0.7$ we find the z-score corresponding to $\hat{p} = 0.75$.

$\hat{p}$	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
z	−2.85	−2.21	−1.581	−0.949	−0.316	0.316	0.949	1.581	2.21	2.85	3.48
$\Phi(z)$	0.002	0.014	0.057	0.171	0.376	0.624	0.829	0.943	0.986	0.998	1.000

d



At the resolution of this graph, the theoretical probability distribution curve and the normal approximation to it are not separated. Look at the tables.

TEXTBOOK HOMEWORK · Exercise 17D

NOTE

In the previous exercise we used continuity corrections when we approximated a discrete binomial distribution by a continuous normal distribution. For large values of n this correction is less necessary. In this lesson, however, using the continuity correction when using sample proportions leads to much messier calculations, and the numbers are mostly large, so it will seldom be applied.

FOUNDATION

1 **a** A student tosses five coins and the results are: H T H T T. What is the sample proportion $\hat{p}$ of heads in this experiment?

b A student selects ten cards from a standard pack with replacement, and records the suit: ♡♣♠♢♢♠♡♠♠♢. What is the sample proportion $\hat{p}$ of spades in this sample?

c A manufacturer takes a sample of 12 items from their recent batch of gizmos, testing each item to see if it passes quality control (P) or not (F). The results were: P P P F F P P P P F P P. What is the proportion $\hat{p}$ of items that pass?

2 A single fair coin is tossed five times, and the number x of heads is recorded.

a Copy and complete the upper table, using the binomial probability formula.

x	0	1	2	3	4	5
$P(X = x)$						

b Copy and complete the lower table to convert part **a** to a table of probabilities of the sample proportions. You will need to divide each score x by 5 to obtain the corresponding sample proportion.

$\hat{p}$						
$P(\hat{p})$						

c Calculate the mean of the second table.

d How would you interpret this mean?

3 Every Saturday for 20 weeks, a marketer surveyed five people chosen at random in a suburban shopping centre. The first question is, 'Do you live in the suburb?', and the weekly frequency of the number x of 'Yes' answers is given in the upper table.

x	0	1	2	3	4	5
f	1	1	3	2	6	7

a Copy the lower table, and complete it to show a table in which the two rows are the sample proportions $\hat{p}$ and the relative frequencies f_r.

$\hat{p}$						
f_r						

b Calculate the mean of this second table.

c What is this mean an estimate of?

4 A coin is tossed 10 times, and a student is asked to calculate the probability that more than 75% of the coins will show heads.

a If more than 75% of the coins show heads, how many coins would this be?

b Use the binomial probability formula to determine the probability of obtaining more than 75% heads. (On this occasion the calculation is easily done without a normal approximation.)

6 Information has been recorded about whether the 32 members of a class buy their lunch regularly at the school canteen:

1. James N	2. Kate N	3. Xavier N	4. Jimmy N
5. Clyde Y	6. Bob N	7. Liam N	8. Agata N
9. Irene N	10. Aqila Y	11. Sonny Y	12. Andrea N
13. Magarida N	14. Terry N	15. Iman N	16. Lucie Y
17. Ping N	18. Maddy Y	19. Kamal Y	20. Xue Y
21. Odette N	22. Billy N	23. Chang N	24. Nahla N
25. Craig Y	26. Jerry N	27. Zahra Y	28. Jun N
29. Nara Y	30. Dakarai Y	31. Lerato N	32. Sahar Y

In order to generate random samples, each student has been given a unique identifying number.

a Calculate the fraction of students who buy their lunch regularly at the canteen. This is called the *population proportion*.

b Kamal generates the five random numbers 12 15 3 30 17, thus generating the sample of 5 students Andrea, Iman, Xavier, Dakarai, Ping. What proportion of these five students buy their lunch regularly at the canteen? This is called a *sample proportion*.

c Copy the table below. Then use the following sets of five random numbers to generate 10 sample proportions for $n = 5$. Enter your results in the table: the first sample was dealt with in part **b** and is already included in the tally. (Notice that repetition is allowed: this is sampling with replacement.)

12 15 3 30 17	3 25 17 17 20	27 7 24 26 2	20 9 21 10 16
26 6 11 5 25	29 24 23 27 3	22 11 25 9 8	27 14 22 11 20
9 28 1 17 1	10 32 24 30 13		

$\hat{p}$	0	0.2	0.4	0.6	0.8	1.0
Tally		|				
Frequency						

d Draw a dot plot for the distribution obtained in part **c**.

DEVELOPMENT

7 In a local election, 20% of the people voted independent. What is the probability that if 500 people are chosen for a random survey, more than 22% of them voted independent? Use a normal approximation to the sample proportion.

8 A card is selected from a standard pack and it is then returned. This experiment is repeated 80 times. What is the probability that a hearts card turns up between 20% and 30% of the time, inclusive? Do this:

a By interpreting it as between 16 to 24 hearts inclusive and using a normal approximation with a continuity correction;

b Using sample proportion without any correction.

9 A farmer knows that a certain type of seed is 70% likely to germinate when planted. He plants 300 seeds at the start of the season. Use the normal approximation for the sample proportion to find the probability that:

a at least 65% will germinate,

b between 65% and 75% will germinate.

10 A medication causes a painful reaction in 5% of users.

a In a group of 100 people, find the probability that:
(i) no one reacts,
(ii) less than two per cent of the people react.

b In a larger study into patients' reactions to this medication, 1000 patients are given the medication (to which 5% are known to have a painful reaction). Researchers find that less than 3% of the patients in this study have a reaction.
(i) Use the normal approximation to the sample proportion to determine the probability of this happening by chance.
(ii) What should the researchers conclude?

11 A student is asked to redo Question 3 from Exercise 17C, but using sample proportions. A barrel contains 600 pink and 400 blue counters. At each stage of an experiment, the barrel is stirred well, then a counter is removed, its colour is noted, and it is returned to the barrel. This experiment is repeated 20 times.

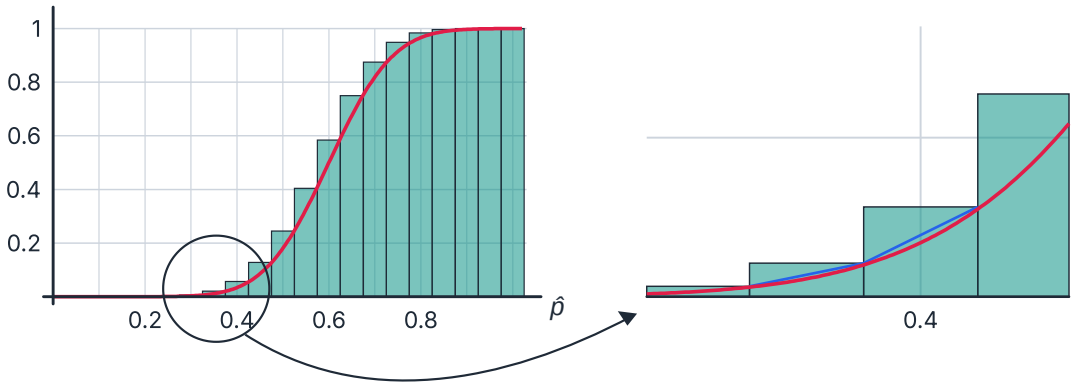
The random variable for this sample proportion is $\hat{p} = \dfrac{X}{n}$, where X is the corresponding binomial random variable with $n = 20$ trials.

The student redraws his cumulative histogram to explore the idea of *sample proportion*. On the horizontal axis, he divides his results by 20 to record the fraction of pink counters occurring in the draw.

a What is the mean and standard deviation of the sample proportion random variable $\hat{p}$?

Recall the formulae $\mu = p$ and $\sigma^2 = \dfrac{pq}{n}$, where $q = 1 - p$.

He overlays the cumulative normal curve with the same mean and standard deviation on top, and finds a very good agreement.



b Explain why the distribution of $\hat{p}$ with values between 0 and 1 is not a continuous distribution.

c To confirm his previous result, he decides to calculate the probability that $\hat{p} \leq 0.4$, meaning that no more than 40% of the counters are pink. Assuming that the distribution is approximately that of a normal random variable $N(\mu, \sigma^2)$, calculate the required probability $P(\hat{p} \leq 0.4)$. Do *not* attempt any continuity correction.

d Calculate the percentage difference in the exact answer 5.7% obtained by a direct binomial calculation. Is there good agreement?

e Calculate the probability that no more than 75% of the counters are pink, that is, $P(\hat{p} \leq 0.75)$.

f Compare the percentage error in this second answer with the exact result 95% and explain why the approximation is better than parts **c**–**d**.

12 In this question we investigate the accuracy of a normal approximation to the sample proportion for various sample sizes n.

A coin is tossed repeatedly and the proportion of heads is recorded. The exact theoretical probability of obtaining at most 52% heads is calculated using the binomial distribution, and recorded in the second row of the table below for differing sample sizes.

n	1000	500	100	50	25
exact	0.9026	0.8262	0.6914	0.6641	0.6550
approx					
% error					

a Use a normal approximation for the sample proportion to fill in the second row of the table.

b In each case calculate the percentage error in the approximation for each of these samples. Record your results in the third row of the table.

c Comment on the accuracy of your approximations for various sample sizes n.

13 A manufacturer distributes tins of pineapple under a recognised brand name, and also as a generic supermarket no-name product. In a trial, 50 customers are given a tin of each and later asked to express a preference. The customers in the trial must choose one product as their favourite. Assuming that there is no difference between the two products, the probability of choosing the branded pineapple should be 0.5. It is found that more than 60% of customers prefer the branded pineapple. Calculate the probability of this using the normal approximation for the sample proportion, then comment.

14 Long-term trials have showed that 30% of patients with a certain disease respond to treatment by a company's drug. In further trials, 100 patients chosen at random from those with the disease are given a higher than usual dosage of the drug, and 40% respond positively. What is the probability that 40% or more could respond positively purely by chance?

15 The variance of the sample proportion $\hat{p}$ for a binomial distribution is $\sigma^2 = \dfrac{pq}{n}$. The variance measures the spread of the distribution of $\hat{p}$ around the population proportion p, thus it is customary to take n sufficiently large to ensure that σ is small.

a Assume that 70% of residents on a college campus are living at home, and that researchers want to choose a sufficiently large sample to mirror this statistic. How big will a sample need to be to ensure that the standard deviation of $\hat{p}$ is less than:
(i) 4% (ii) 3% (iii) 2% (iv) 1% (v) $k\%$

b Repeat this question if a new survey finds that the number of residents living at home is 80%.

c Repeat for $p = 50\%$.

16 [Simulation]

a Ten coins are tossed.
(i) What is the probability of getting exactly six heads?
(ii) If this experiment is carried out 40 times, use your answer to part (i) to predict how many times you expect to get exactly 6 heads.

ENRICHMENT

17 Two dice are thrown and success is recorded if the sum is at least 10.

a Find the probability of success.

b Use the exact binomial distribution to find the probability of at most four successes on 20 throws.

c Use a normal approximation to the binomial to estimate the probability of at most four successes without continuity correction.

d Repeat part **c** with continuity correction.

e Use sample proportion to estimate the probability of at most 20% successes. Do not use continuity correction. Explain why your result agrees with part **c**.

18 [Confidence intervals]

a Let $X \sim N(\mu, \sigma^2)$.

(i) Show that 95% of the data lie within 1.96σ of the mean μ, that is,

$$P(\mu - 1.96\sigma < X < \mu + 1.96\sigma) \doteq 95\%.$$

(ii) We interpret this to mean that a randomly chosen value X will lie within the interval $[\mu - 1.96\sigma,\ \mu + 1.96\sigma]$ ninety-five per cent of the time. Draw a diagram to show that for a given value x of the random variable, μ will lie within the interval $[x - 1.96\sigma,\ x + 1.96\sigma]$ ninety-five per cent of the time.

b In a sample of 100 school pupils, 67 surf the internet more than 7 hours a week. Thus we might estimate that 67% of all school pupils do the same, but how confident can we be of this estimate?

(i) Estimate the population standard deviation using this single sample, from the formula $\sigma^2 = p(1-p)/n$, where we use the estimate $p \doteq \hat{p} = 67\%$.

(ii) Find the margin of error 1.96σ and find the 95% confidence interval, which is $[\hat{p} - 1.96\sigma,\ \hat{p} + 1.96\sigma]$, indicating that 95% of the time we take such a sample, the population proportion will fall within this interval.

(iii) Using the estimate $p \doteq 67\%$, how large does the sample size n need to be to reduce the margin of error to 1%?

PAST-PAPER HOMEWORK · mini exam

Suggested time: 30 minutes **19 marks**

Calculator allowed · use the standard normal table at the end of the Revision section · show working · hints are below each question if you get stuck

Where a question does not say whether to use a continuity correction, the worked solutions follow the school's own method, which uses no continuity correction, and say so.

PP 1 **Baulkham Hills High School, 2020, Year 12 Trial Examination, Mathematics Extension 1, Q4** **1 mark**

In a large population, the probability that a person is left-handed is 0.3. Let $\hat{p}$ be the random variable that represents the sample proportion of left-handed people from a sample size of n, drawn from the population.

What is the smallest value of n such that the standard deviation of $\hat{p}$ is less than or equal to 0.07?

A. $n = 42$ B. $n = 43$ C. $n = 61$ D. $n = 62$

Hint: which distribution does $\hat{p}$ have here?

PP 2 **Killara High School, 2021, Year 12 Mathematics Extension 1 Trial Examination, Q12(a)(ii)** **2 marks**

An online furniture store claims that 90% of all orders are shipped within 54 hours of a customer placing an order through the store's website. Ty ordered a total of 150 various pieces of furniture from the store for his company.

(ii) Show that the expected value and the standard deviation of the sampling proportion are 0.9 and 0.0245 respectively. 2

Hint: what are n and p here?

PP 3 **James Ruse Agricultural High School, 2023, Year 12 Trial HSC Examination, Mathematics Extension 1, Q15(c)** 4 marks

It is known that the probability of obtaining a head when tossing a particular biased coin is 0.2. A random 100 flips of this biased coin is conducted and the number of heads are observed.

(i) Justify why the proportions of heads can be approximated using the normal distribution. 1

(ii) Determine the mean and standard deviation of the proportion of heads. 1

(iii) Hence or otherwise, approximate the probability that the proportion of heads obtained will be between 0.1 and 0.3 inclusive, to 4 decimal places. 2

Hint: what are n and p here?

PP 4 **Hurlstone Agricultural High School, 2023, Higher School Certificate Trial Examination, Mathematics Extension 1, Q13(c)(ii)** 4 marks

In a certain school, 23% of Year 12 students study Mathematics Extension 1.

(ii) The Principal meets with a random sample of 60 Year 12 students, to discuss their Year 12 HSC courses. 4

What is the probability that more than 30% of the students that meet with the Principal study Mathematics Extension 1?

(You may assume that the sample of students is approximately normally distributed, and make reference to the extract below from a table of z scores).

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177

Hint: what are n and p for this sample?

PP 5 **Killara High School, 2024, Year 12 Examination, Mathematics Extension 1 HSC Trial Examination, Q14(c)** **5 marks**

A cereal company is interested to know whether people drink the cereal milk. They asked a random sample of 1012 Australian adults what they do with the milk in the bowl after they have eaten the cereal. Of the respondents, 67% said that they drank it. Suppose that 70% of Australian adults actually drink the cereal milk.
Let $\hat{p}$ be the proportion of people in the sample who drink the cereal milk.

(i) What is the mean and standard deviation of the sampling distribution of $\hat{p}$? 2

(ii) By calculating the appropriate probability, do you have any doubts about the result of this poll? Justify your solution. 3

Hint: what does the poll tell you about $\hat{p}$?

PP 6 **Abbotsleigh, 2024, Higher School Certificate Assessment 4 Trial Examination, Mathematics Extension 1, Q14(c)(i)** 3 marks

You may use the information [in the standard normal table at the end of the Revision section] to answer this question.

A binomial variable, X, is known to have a probability of success, p, of close to 0.81, but the exact value is unknown.

(i) A mathematician needs to estimate the value of p to within one-thousandth of the true result, with 99.7% certainty. 3

To find a more accurate estimate of p, the mathematician plans to perform n trials, and find $\hat{p}$.

What is the minimum number of trials the mathematician needs to perform so that $\hat{p}$ is accurate to within one-thousandth of the true value of p, with 99.7% certainty?

Hint: what does the required certainty tell you about $\hat{p}$?

CHAPTER 17 REVIEW

Chapter review exercise

Mixed practice from the whole chapter, Lessons 17A to 17D. Use the standard normal table at the end of the Revision section.

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1 A marksman finds that on average he hits the target five times out of six. Assuming that successive shots are independent events, find the probability that in four shots:

a he has exactly three hits, **b** he has exactly two misses.

2 Five out of six people surveyed think that Tasmania is the most beautiful state in Australia. What is the probability that in a group of 15 randomly selected people, at least 13 of them think that Tasmania is the most beautiful state in Australia?

3 There are ten questions in a multiple-choice test, and each question has five possible answers, only one of which is correct. What is the probability of answering exactly seven questions correctly by chance alone? Give your answer correct to three significant figures.

4 A card is selected from a pack, its suit is noted, and it is returned. How many times must this be done so that the probability of a heart is more than 95%?

5 An eight-sided die is inscribed with the digits 1–8.

a What is the probability of obtaining an 8 when the die is thrown?

b Six eight-sided dice are thrown. Construct a table for the distribution of the random variable X that counts the number of eights that occur. Record your results correct to 4 decimal places.

c A player needs to get exactly three eights in order to win. How often would you predict this to occur in 1000 throws of the six dice?

d Repeat part **c** if he needs a throw of three or more eights.

6 Are the following experiments Bernoulli trials? If so, state the probability of success p and failure q.

a A coin is tossed, and it is noted if the result is heads or tails.

b Two dice are thrown, and the player wins if the sum is more than 10.

c Tests show that 4 out of every 1000 items pass quality control. Consider the random variable 'number of passes' where an item is selected at random from the manufacturing process.

d A card is drawn from a pack, and its suit is noted.

7 The binomial distribution $B(n, p)$ consists of n independent Bernoulli trials. Find the mean, variance and standard deviation for each distribution.

a $B(20, 0.2)$ b $B(70, 0.5)$ c $B(6, 0.8)$

d $B(120, 0.4)$ e $B(300, 0.1)$ f $B(5, 0.25)$

8 A company manufactures mobile phone cases using a mixture of machinery and traditional techniques. Data shows that the probability that a random case will fail quality control is 5%. An inspector selects a random batch of 60 cases from the warehouse. Let X be the binomial random variable of the number of cases that do not pass inspection.

a What is the mean, variance and standard deviation for this distribution?

b Find the probability that the number of cases that fail to pass lies within one standard deviation of the mean.

c New company standards insist that the number of failures in the batch must be no more than one standard deviation above the mean. Batches that fail to meet this standard are rejected. What is the probability of this?

d Due to the new regulations and the number of rejected batches, the company improves its manufacturing process so that the new experimental probability of failure is reduced to 2%. Repeat part **c** to find the new probability that a batch will be rejected.

9 A coin is tossed 80 times.

a Use the exact binomial formula to find the probability of 38, 39 or 40 heads.

b What is the mean and standard deviation for this binomial distribution?

c Calculate np and nq, and state whether this is a situation where a normal approximation may be used.

d In your own words, explain why we calculate $P(37.5 \leq X \leq 40.5)$ rather than $P(38 \leq X \leq 40)$.

e Find the probability of 38, 39 or 40 heads using a normal approximation.

f What is the percentage error in this normal approximation?

g Clearly in the example above, there was no need to use an approximation, because the probability could be calculated directly. Calculate now the probability of at least 50 heads using a normal approximation (but do not estimate the percentage error).

10 The sum of two dice is recorded.

a Find the probability that the sum is at least 10.

b Use a normal approximation to find the probability that the sum is at least 10 in more than 14 out of 80 throws.

11 A sample of 100 voters are asked whether they intend to vote for the Working Together Party at the next election. Thirty-five respond in the affirmative.

a What is the corresponding sample proportion?

b If a further sample is taken, would you expect the same number of respondents to indicate yes?

c What type of distribution will result if further such samples are taken and the sample proportions are recorded?

12 A bag contains 3 red balls and 2 white balls. Five balls are selected in turn, with replacement, and the number of red balls is recorded.

a Construct a table showing the theoretical sample proportions of red balls that are selected.

b Find the probability that the proportion of red balls is less than:
(i) 40%, (ii) 50%.

c Find the mean, variance and standard deviation for the random variable $\hat{p}$ tabulated in part **a**.

13 A fair die is to be thrown 500 times.

a Find the mean and standard deviation for the sample proportion of sixes in the theoretical distribution for this experiment.

b In one sample of 500 throws, the number of sixes was 70. How many standard deviations is this result below the mean?

14 Long-term records show that the percentage of male babies born in a large hospital is 53%. A study is carried out on the effect of a high potassium diet (white beans, salmon, avocados, almonds, apples and mushrooms) on increasing the probability that the baby will be male. In the group of 653 births under this study, with the mother following this diet, more than 54% were male. What is the probability of this happening by chance? Use a normal approximation for the sample proportion with no continuity correction.